

notebook

August 27, 2025

1 To do

- ~~Analyze lick data~~
 - ~~RT~~
 - ~~Lick rate~~
 - ~~HLI~~
- History kernels
- Add HMM

2 Analyses of drug data from batch #6

3 Load behavioral data

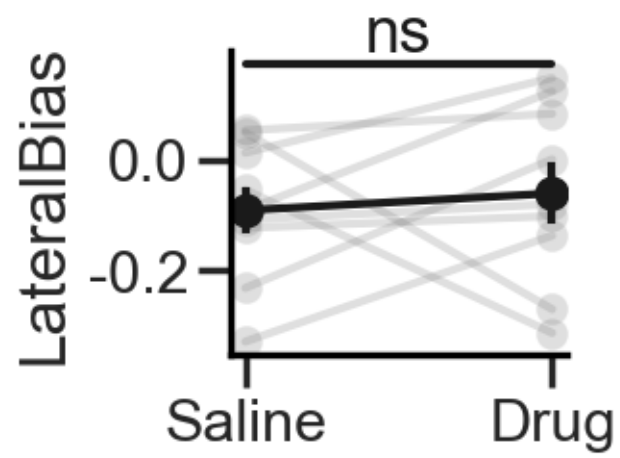
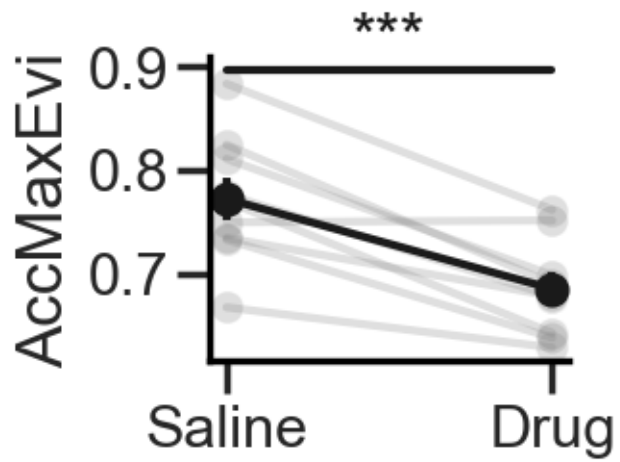
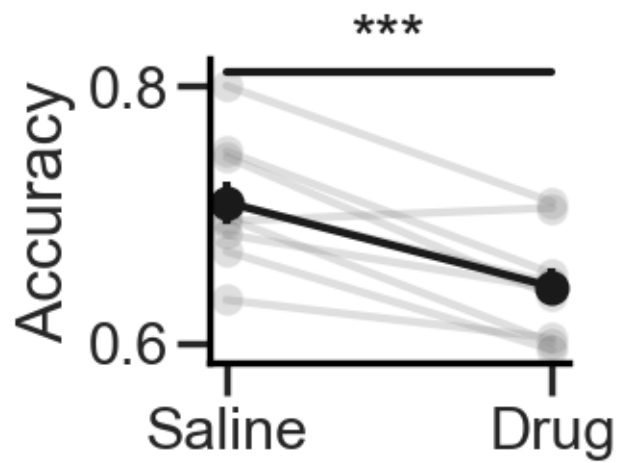
`glue_animals_intersessions` took 0.16 seconds to run

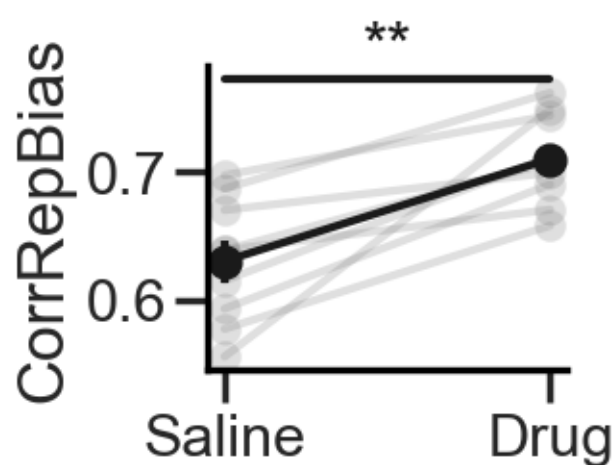
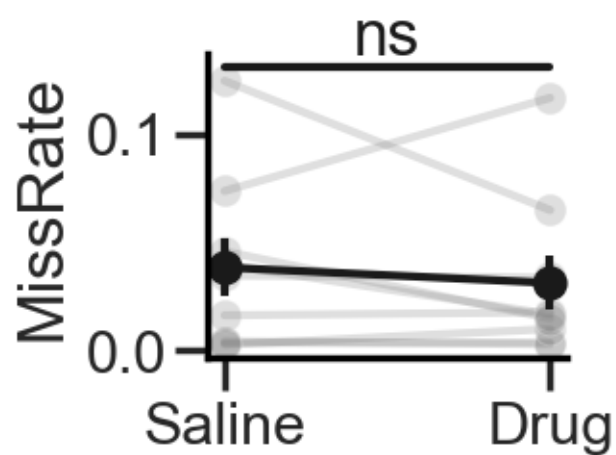
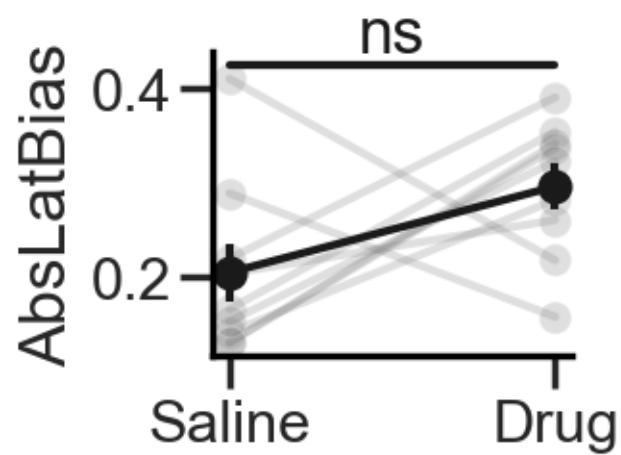
4 Filter data

5 Behavioral metrics

5.1 Standard

Accuracy: t-statistic = 5.083, p-value = 0.001
AccMaxEvi: t-statistic = 5.649, p-value = 0.000
LateralBias: t-statistic = -0.461, p-value = 0.657
AbsLatBias: t-statistic = -1.808, p-value = 0.108
MissRate: t-statistic = 0.751, p-value = 0.474
CorrRepBias: t-statistic = -4.850, p-value = 0.001





5.2 Licks

5.2.1 Add lick data

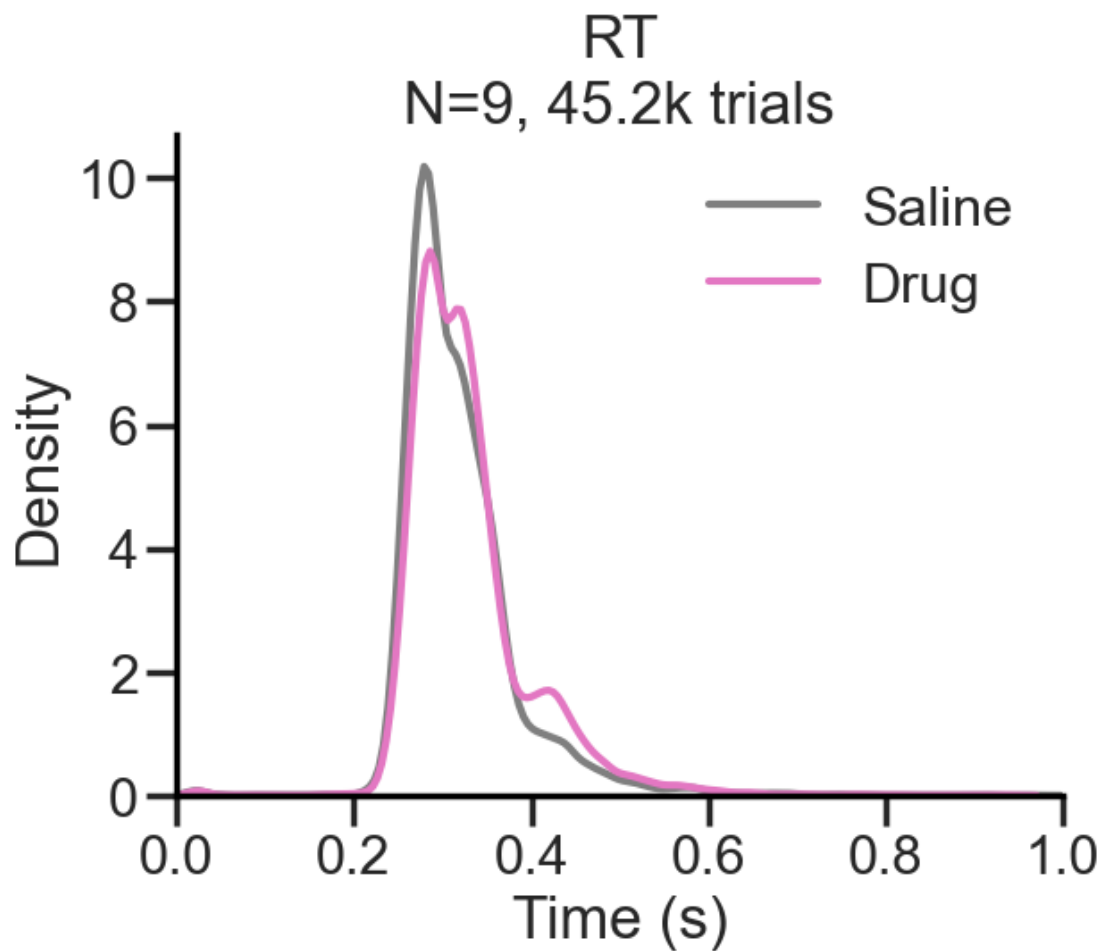
1.11% of premature lick trials (before response window)

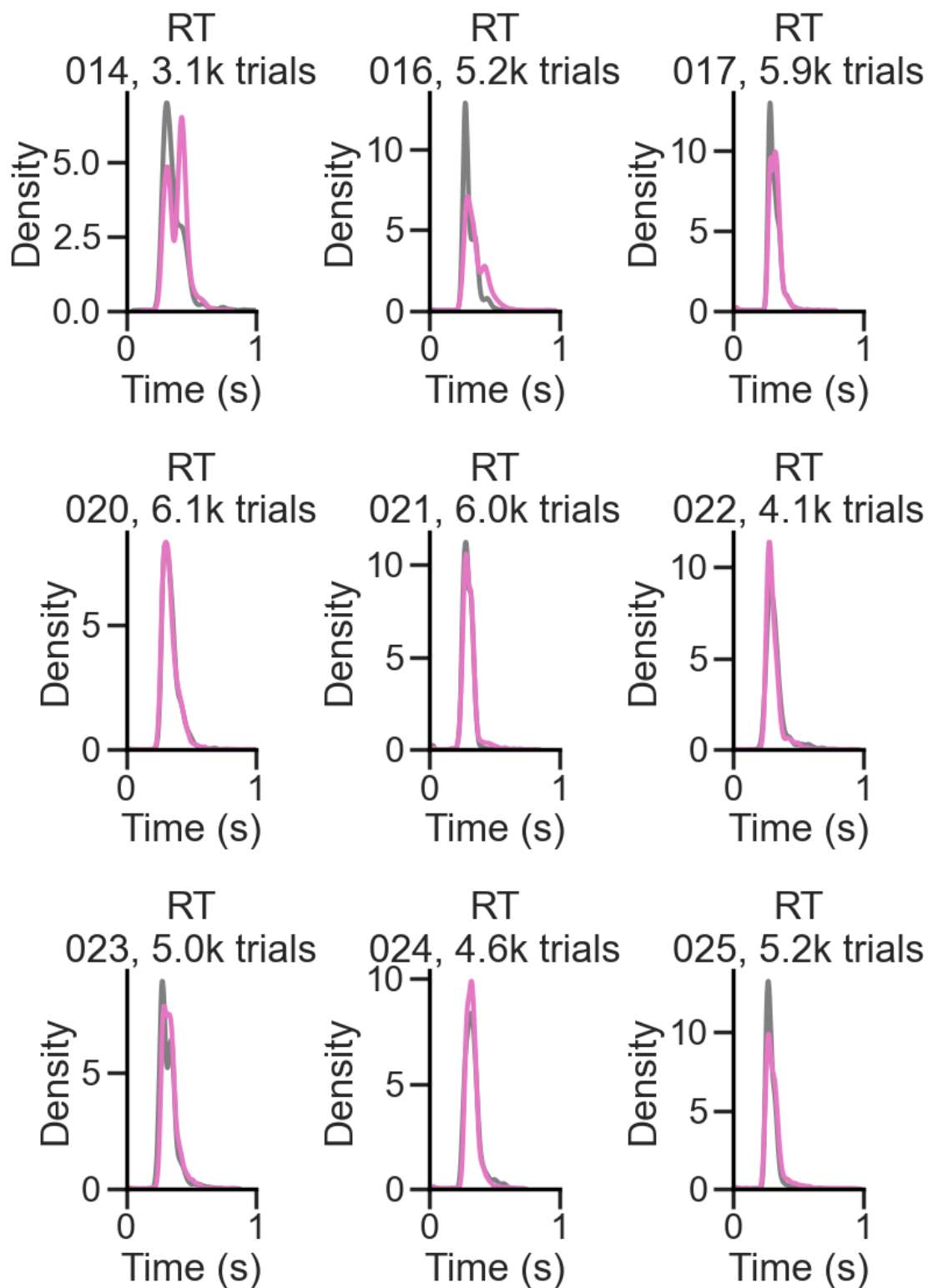
Port1In or Port2In are already lists

```
['absILD', 'LicksLeft', 'LicksRight', 'nLicksLeft', 'nLicksRight', 'nLicks',  
'leftILI', 'rightILI', 'ILI', 'RT', 'RT2', 'SessionHalf']
```

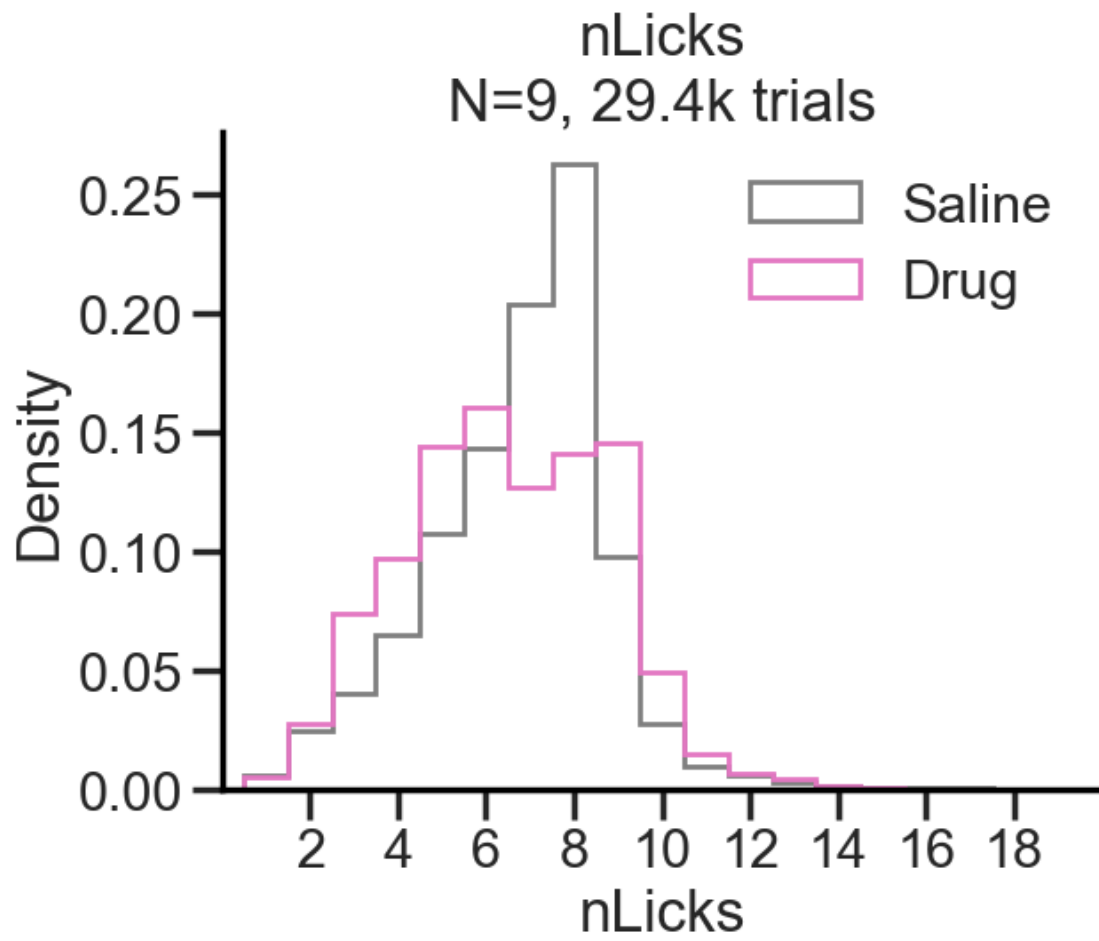
184

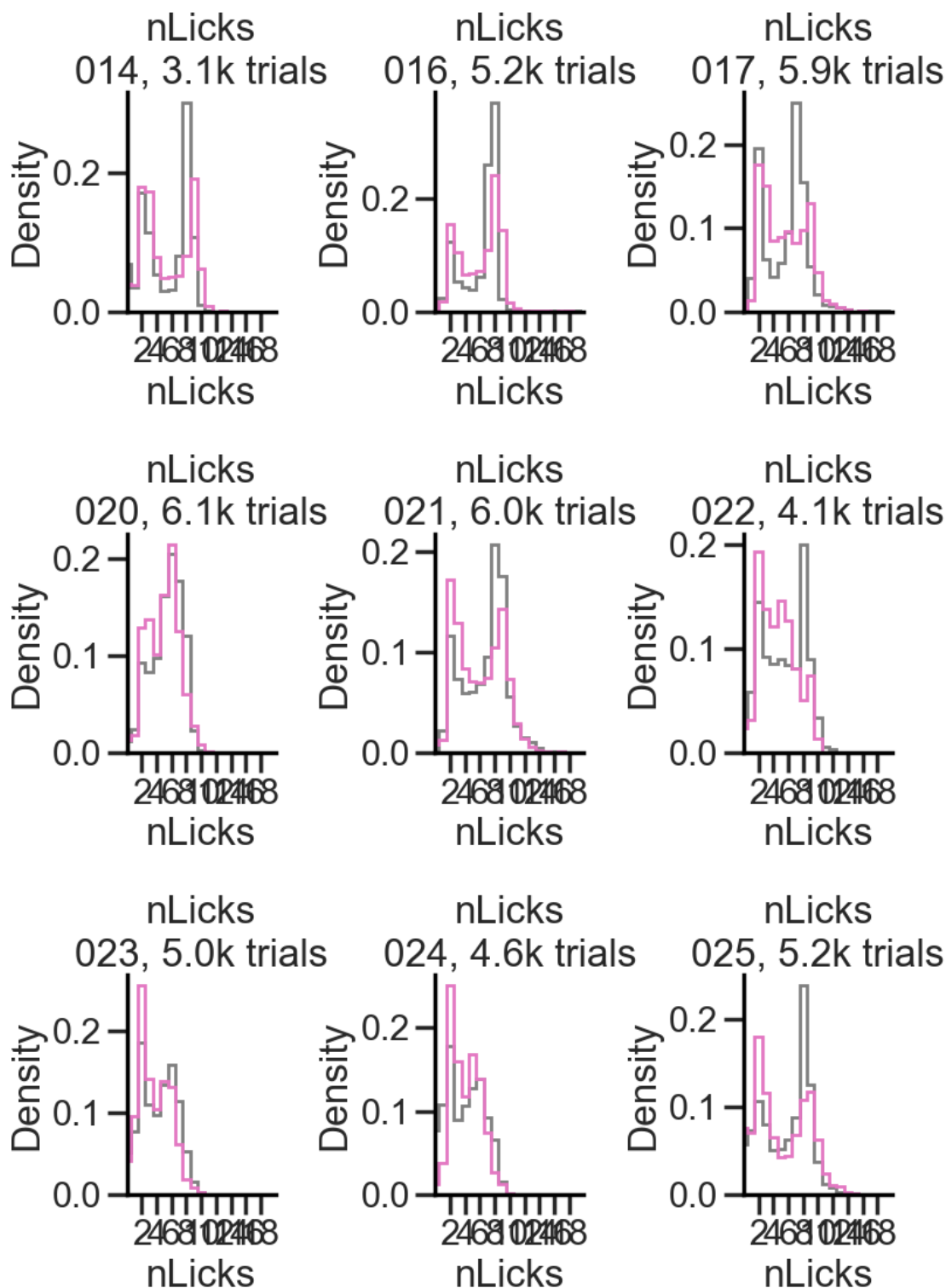
5.2.2 RTs



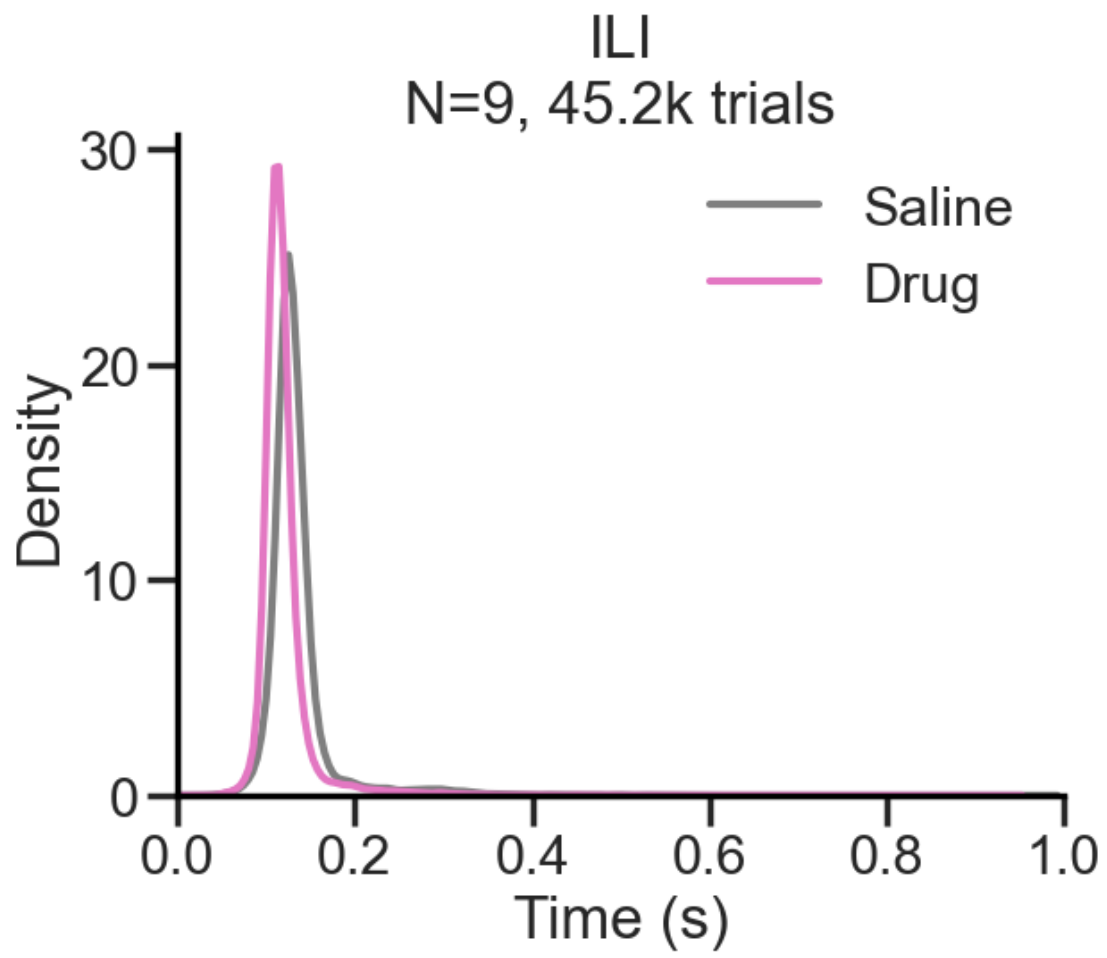


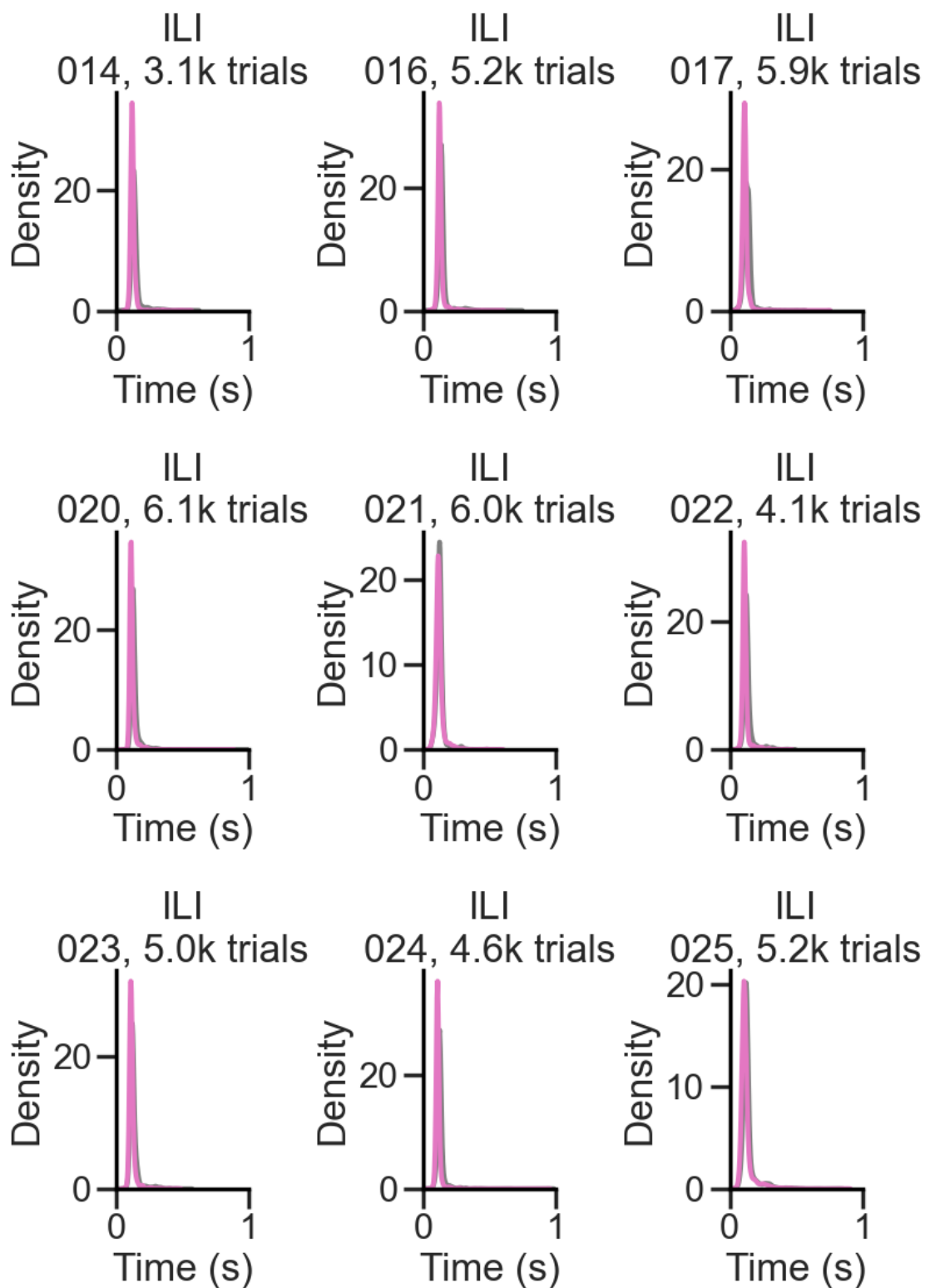
5.2.3 Lick rate





5.2.4 ILI





6 Psychometrics

6.1 Probability choose RIGHT

6.1.1 Single animals

Animal 014

Animal 016

Animal 017

Animal 020

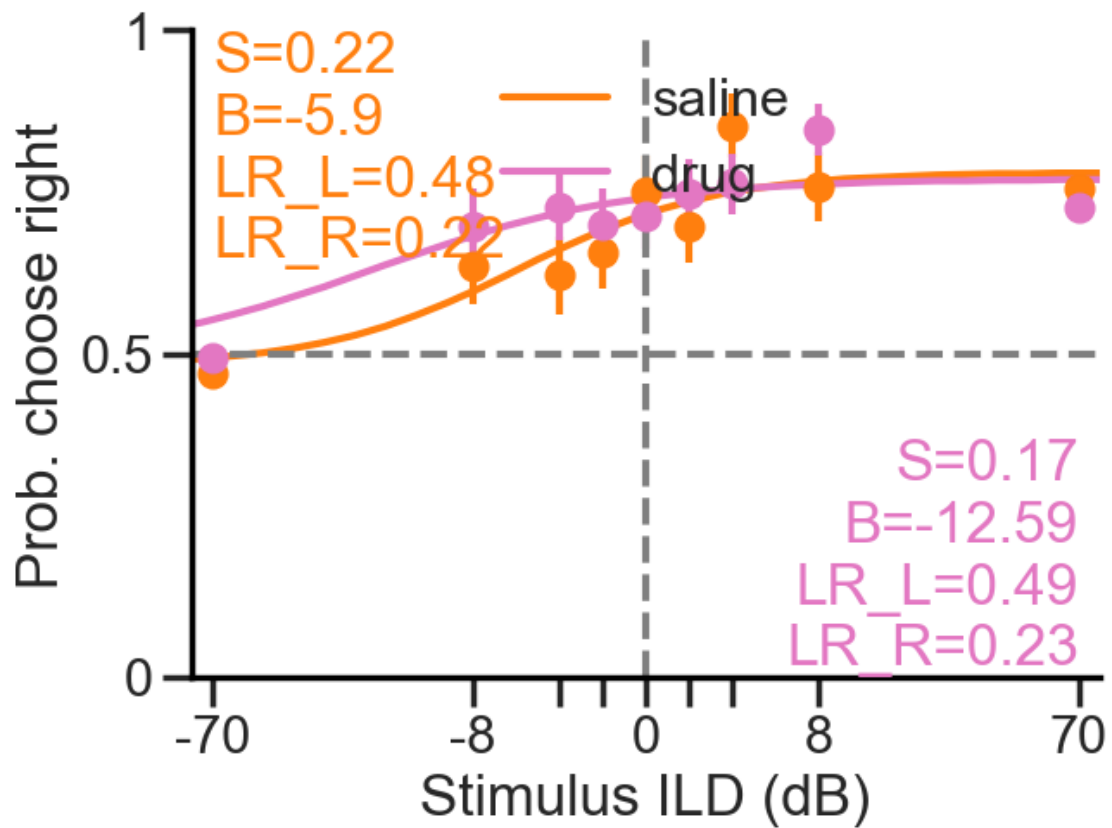
Animal 021

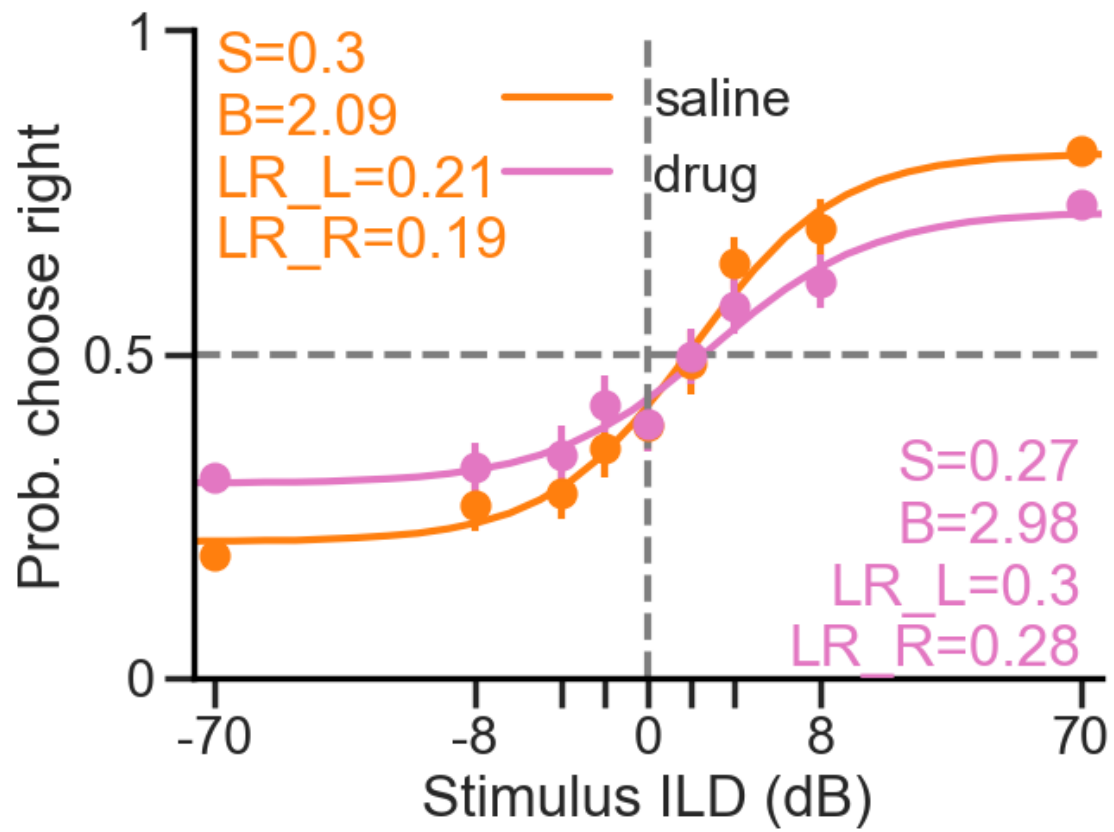
Animal 022

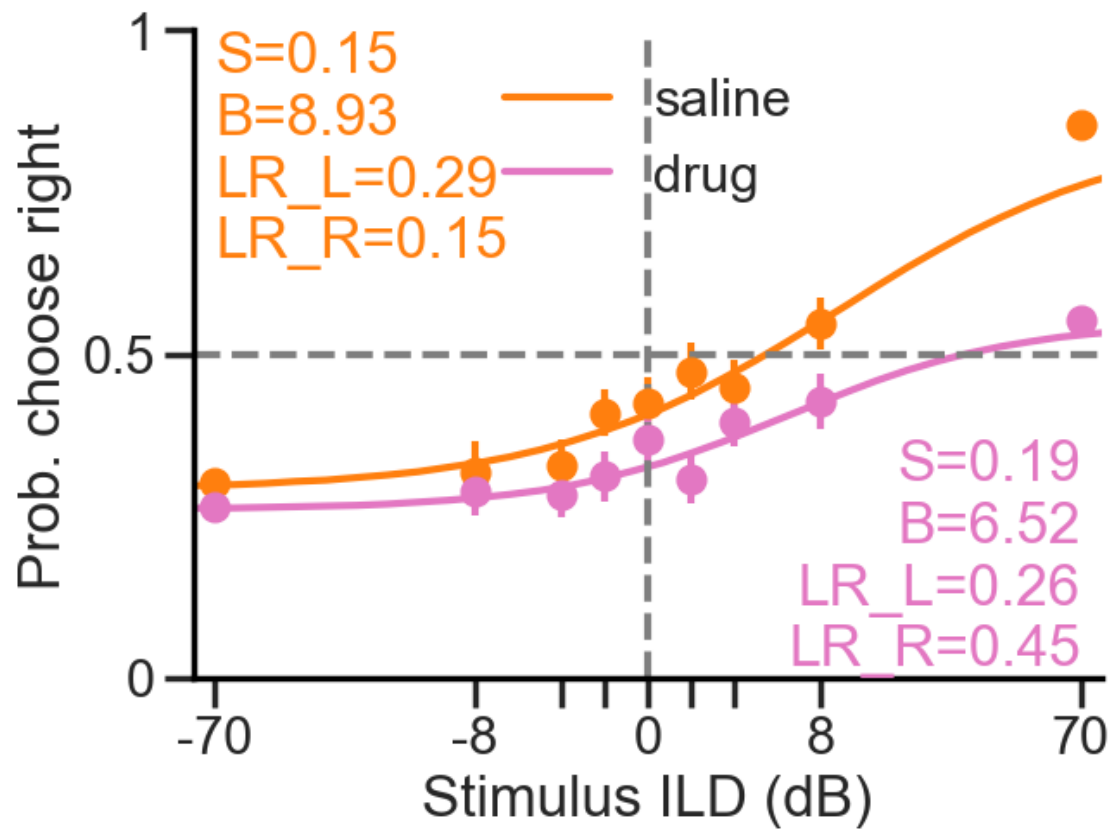
Animal 023

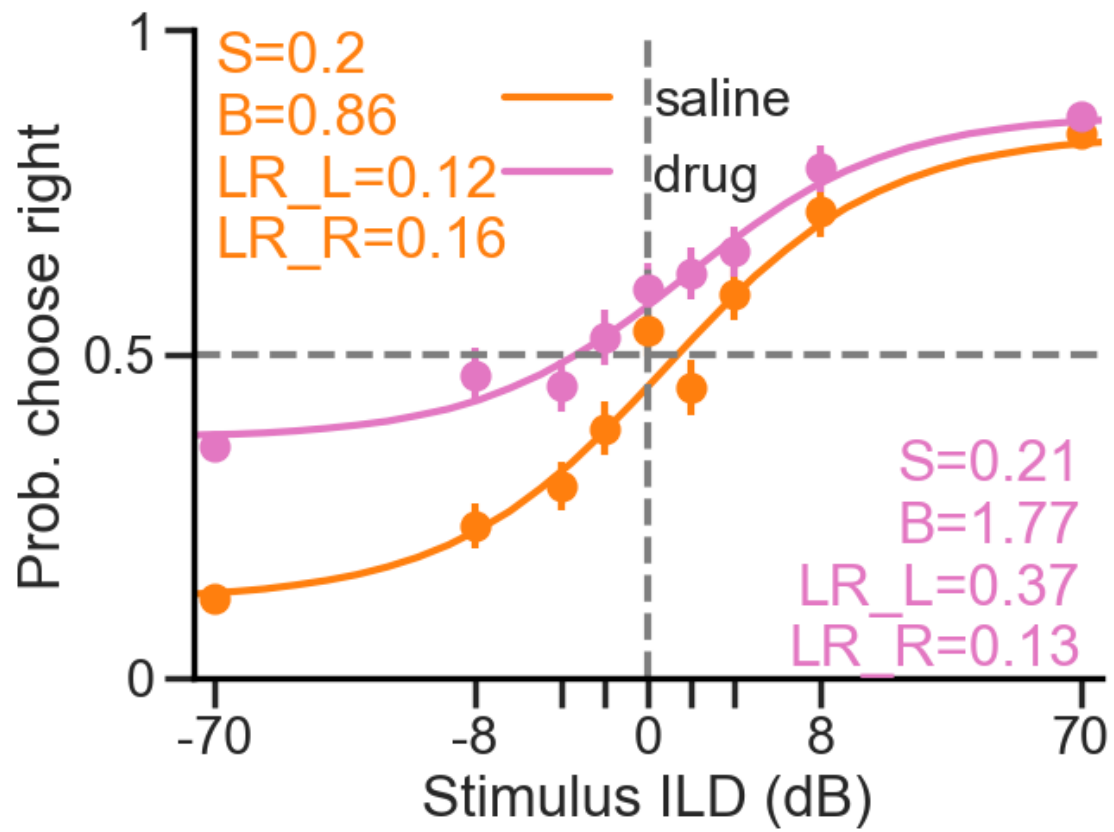
Animal 024

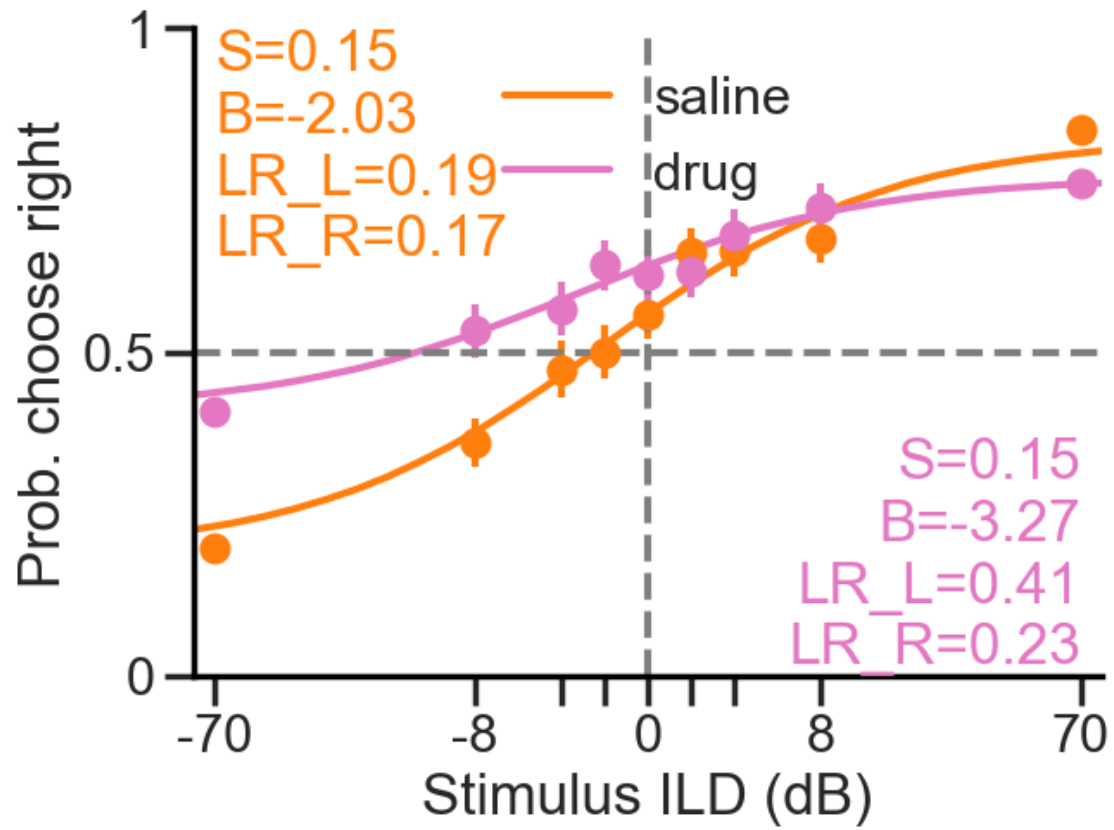
Animal 025

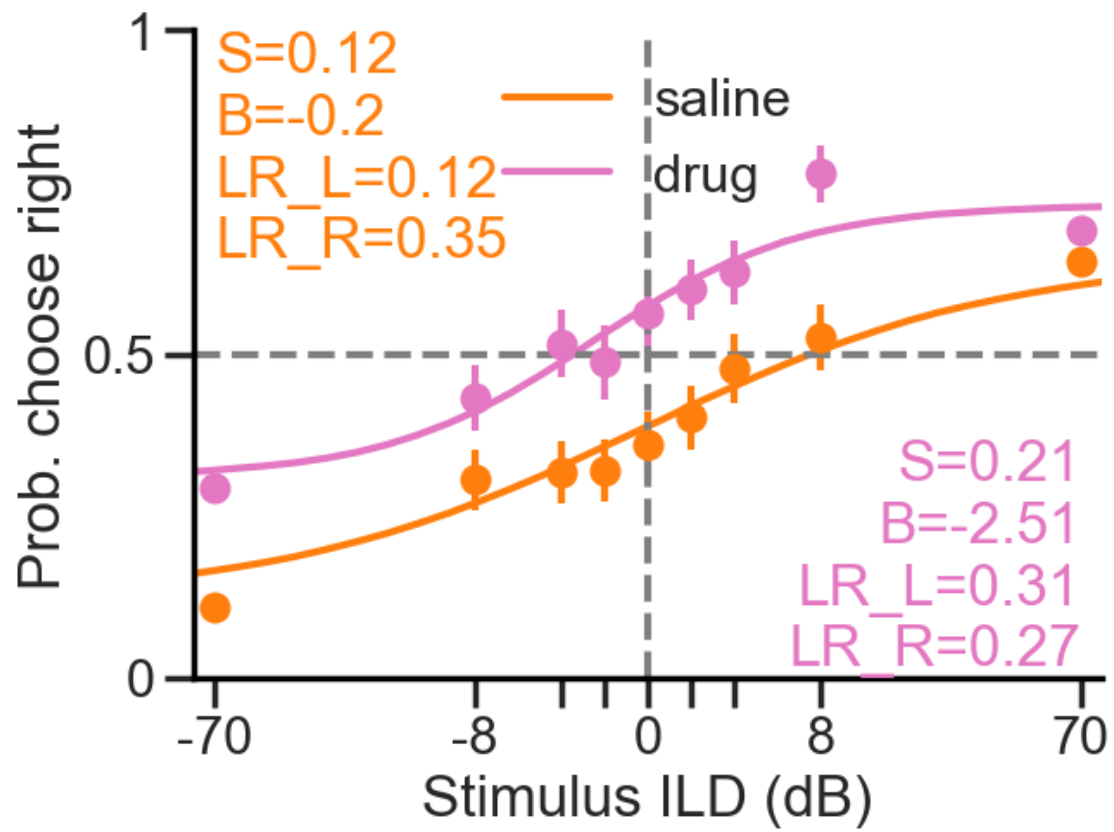


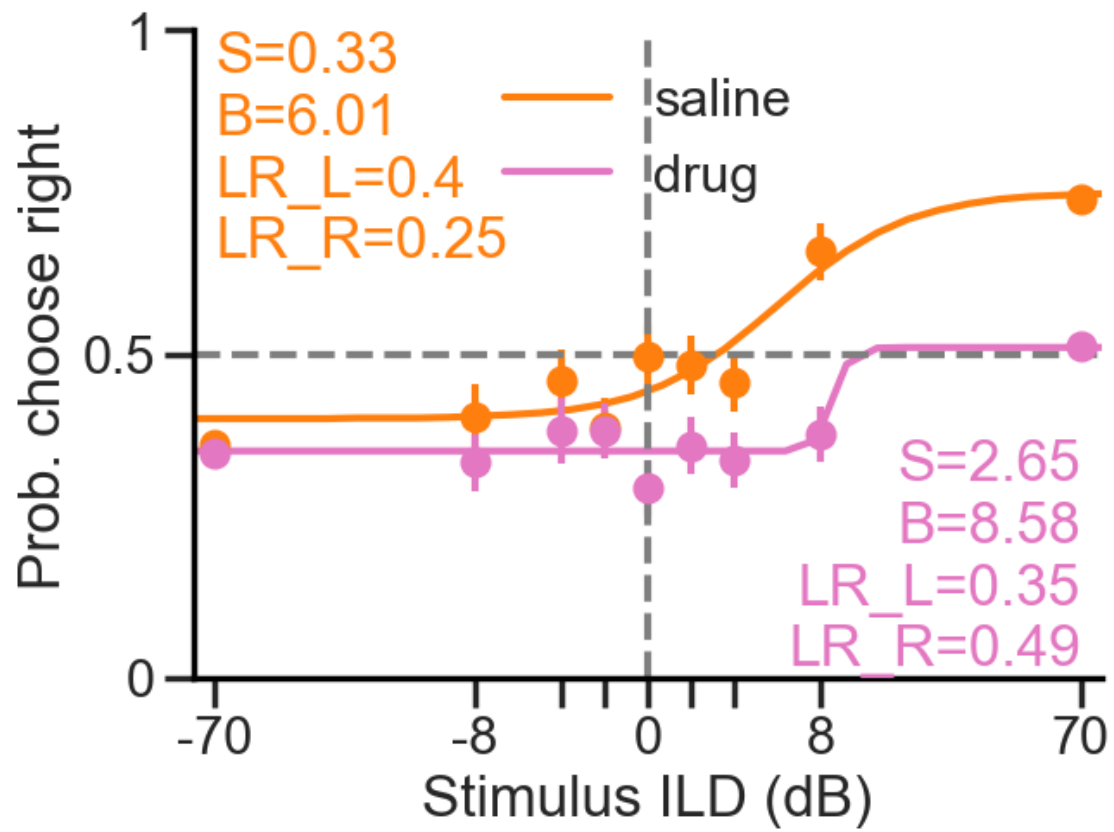


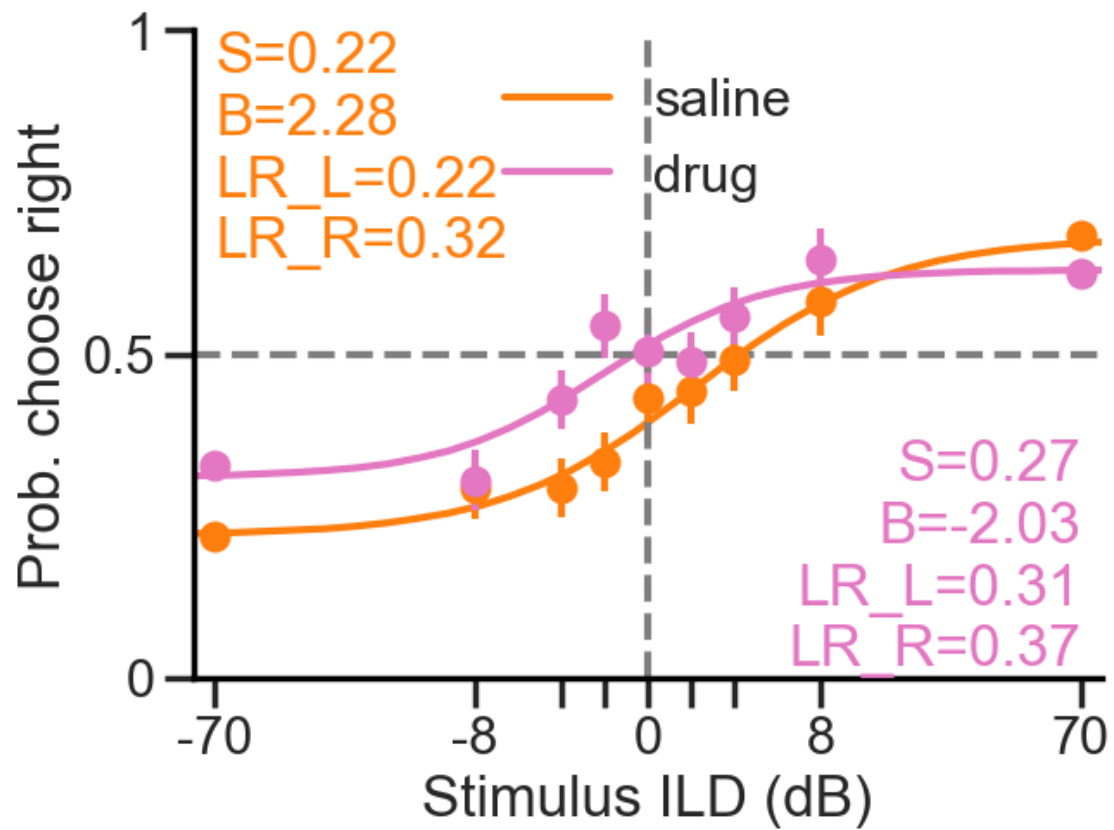


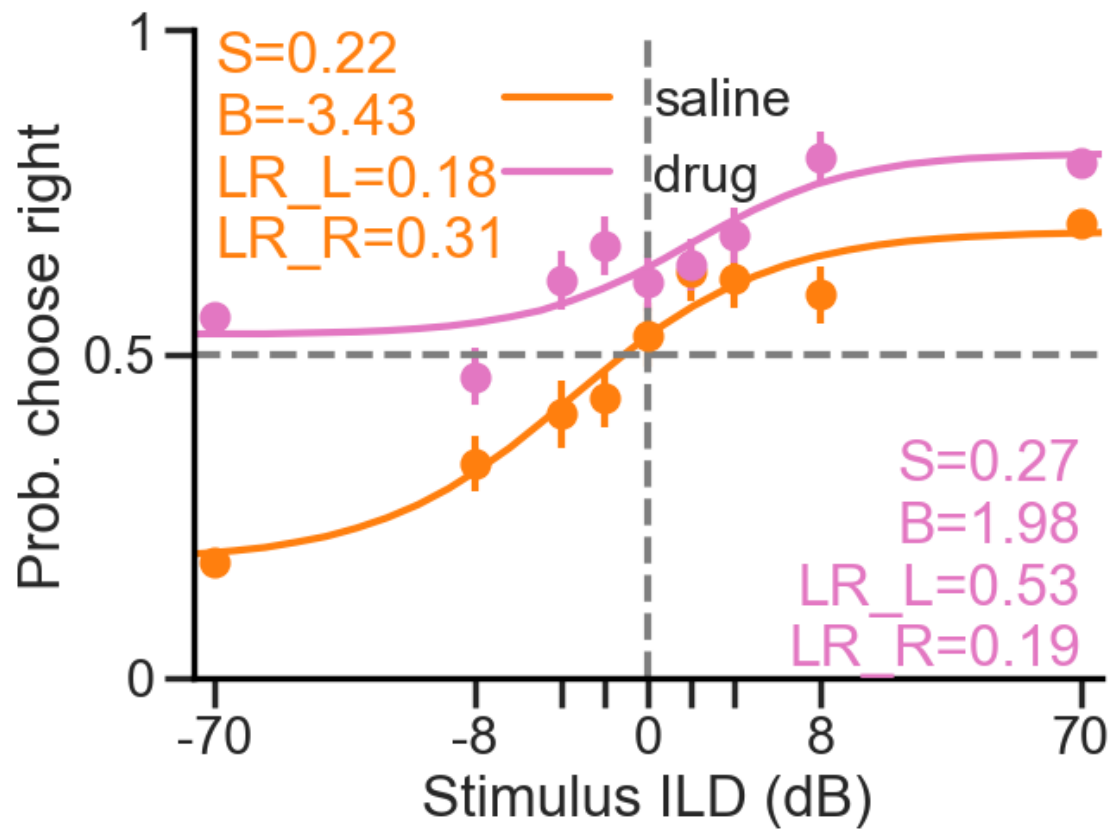




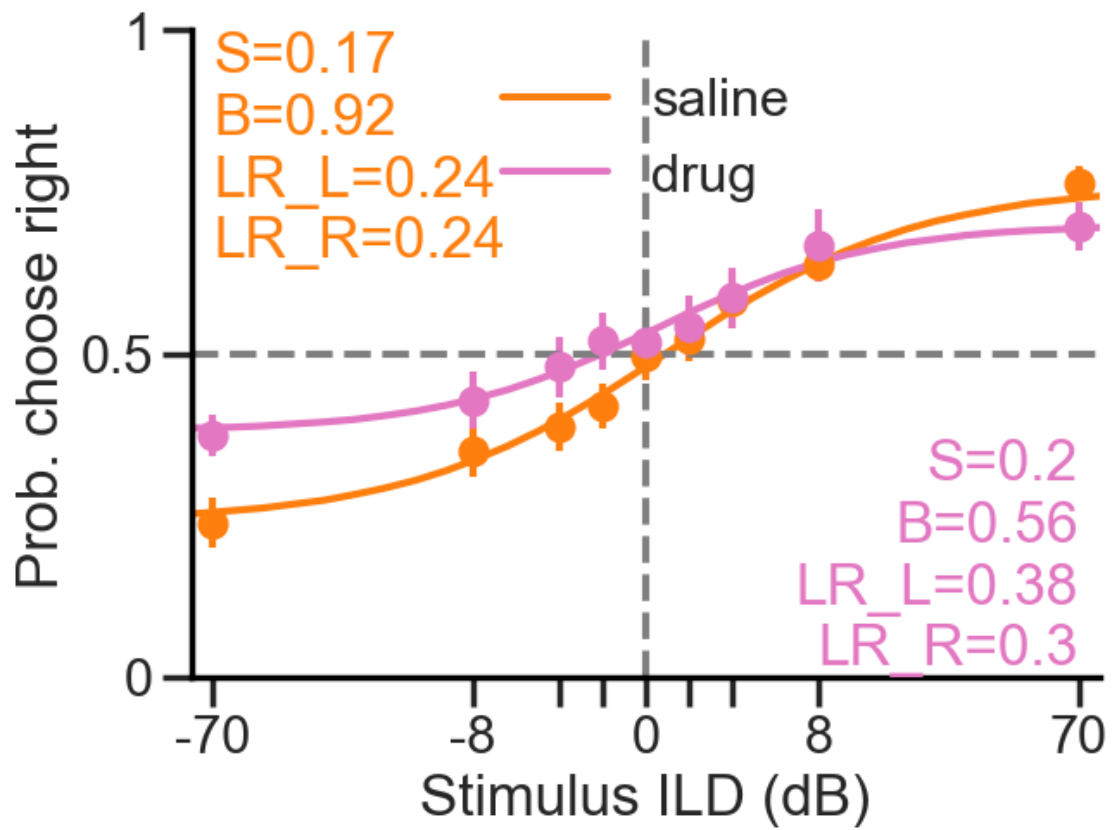






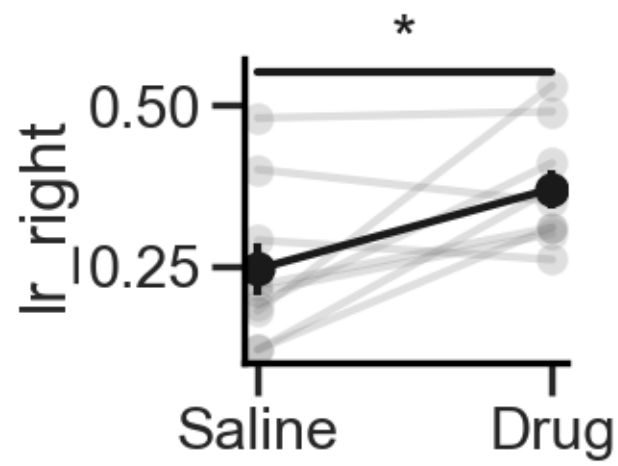
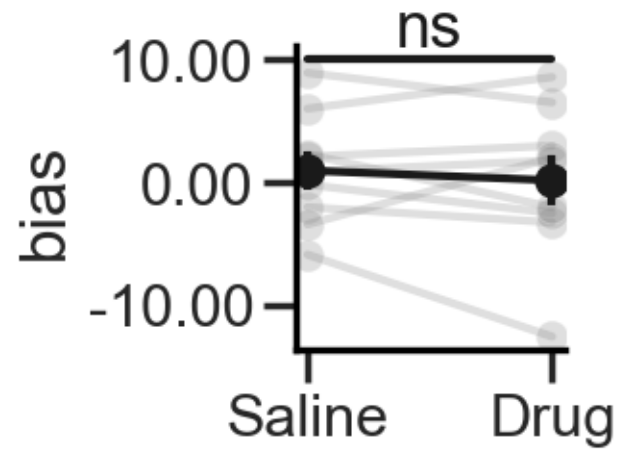
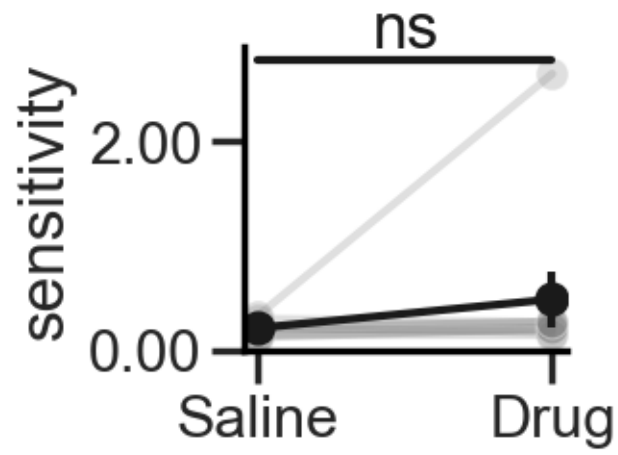


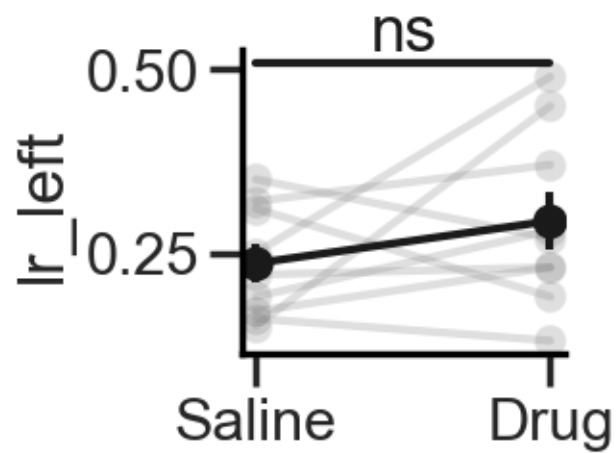
6.1.2 Mean across animals



Stats

sensitivity: t-statistic = -1.077, p-value = 0.313
bias: t-statistic = 0.652, p-value = 0.533
lr_right: t-statistic = -2.728, p-value = 0.026
lr_left: t-statistic = -1.251, p-value = 0.246

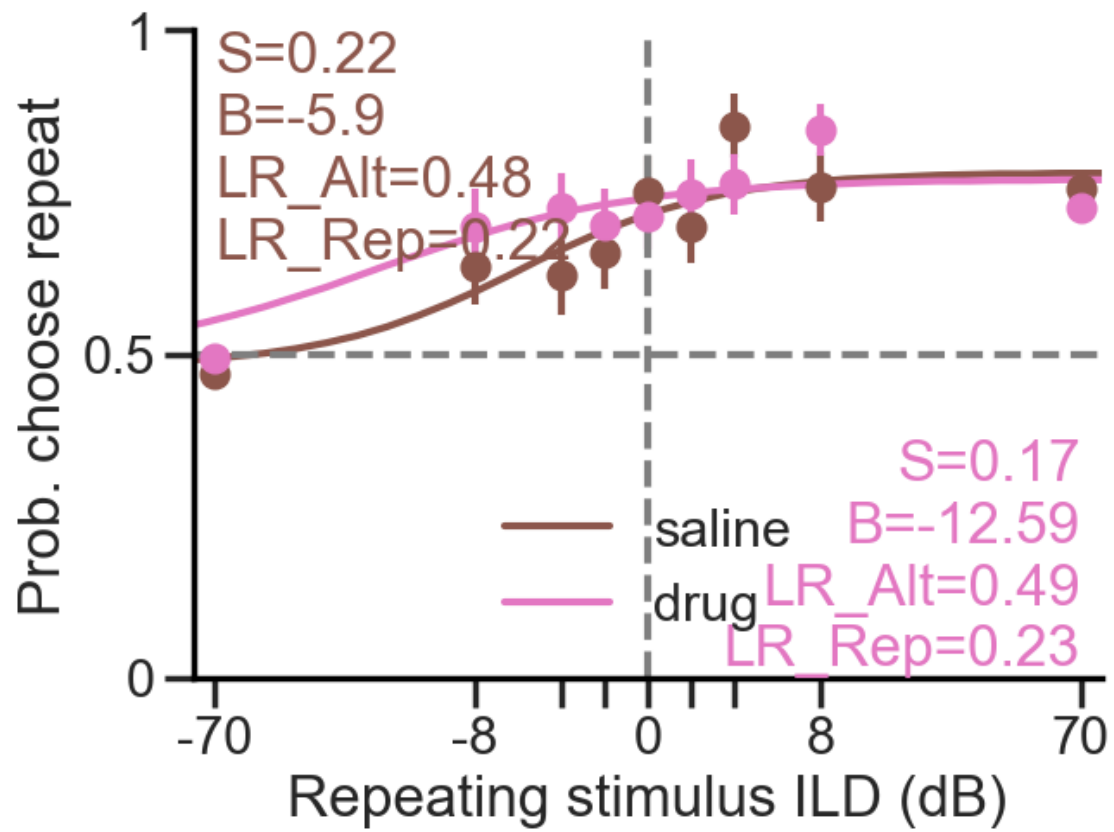


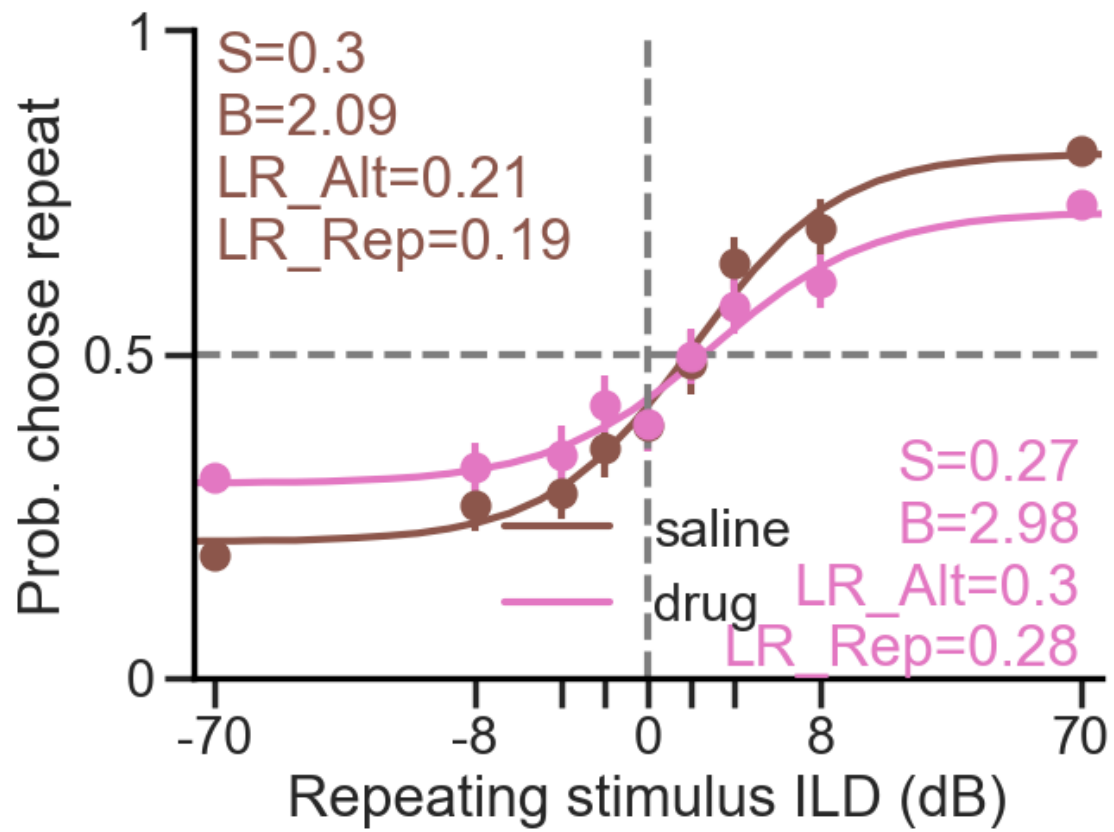


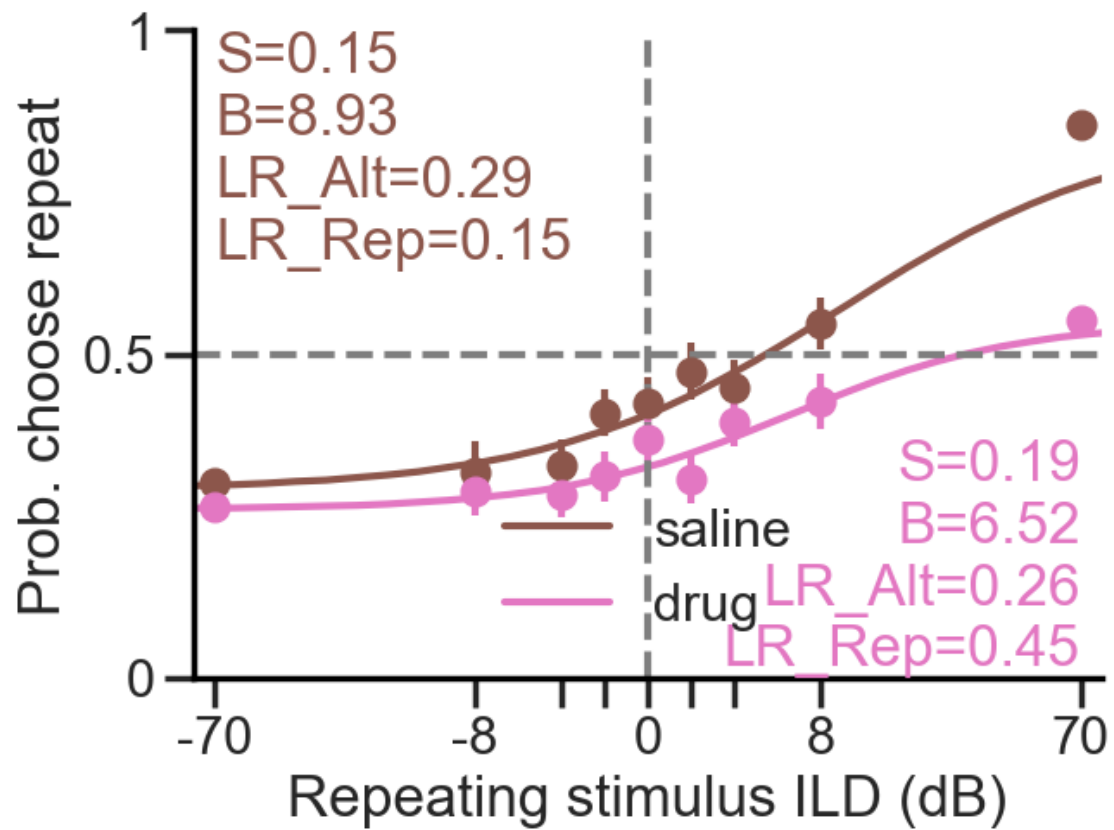
6.2 Probability choose REPEAT

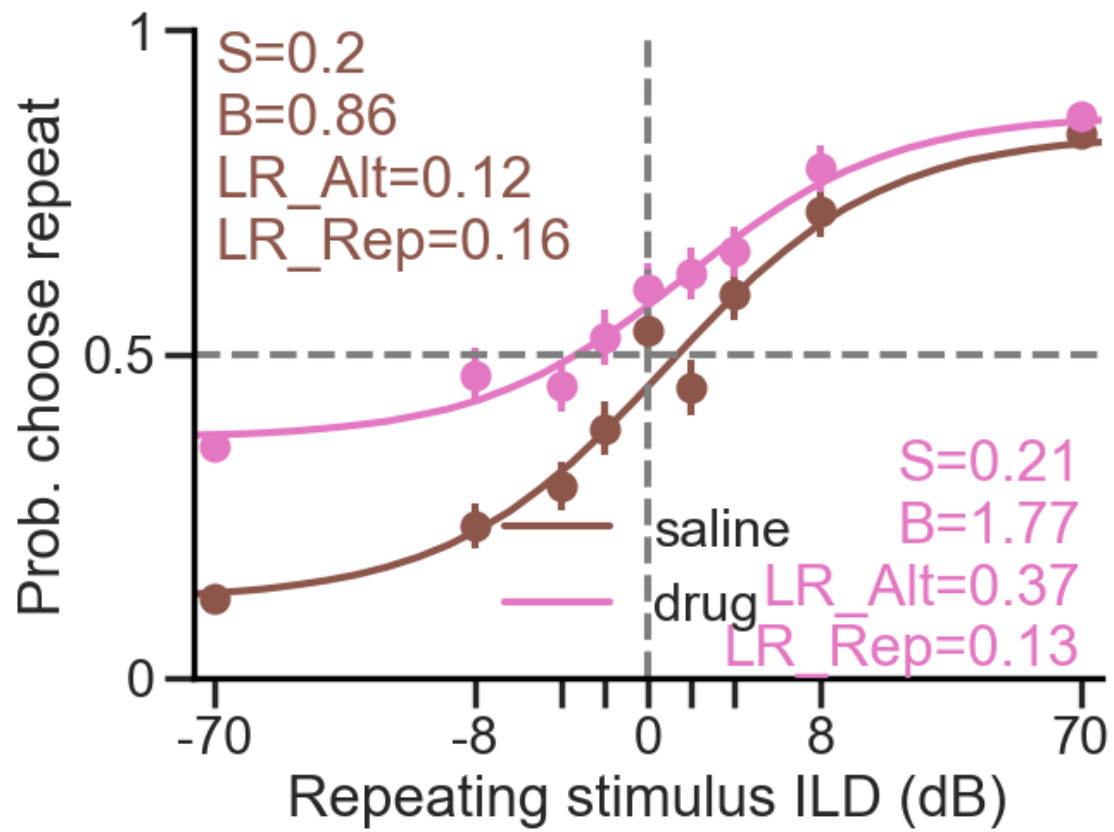
6.2.1 Single animals

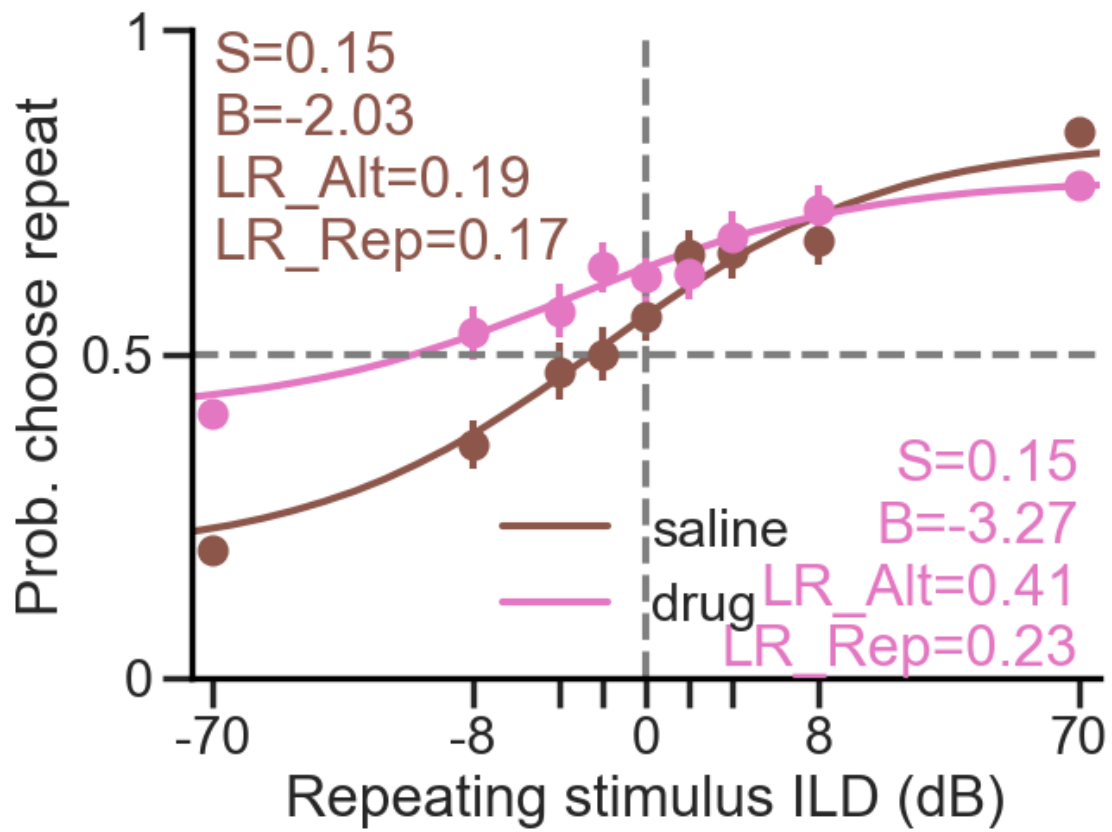
Animal 014
 Animal 016
 Animal 017
 Animal 020
 Animal 021
 Animal 022
 Animal 023
 Animal 024
 Animal 025

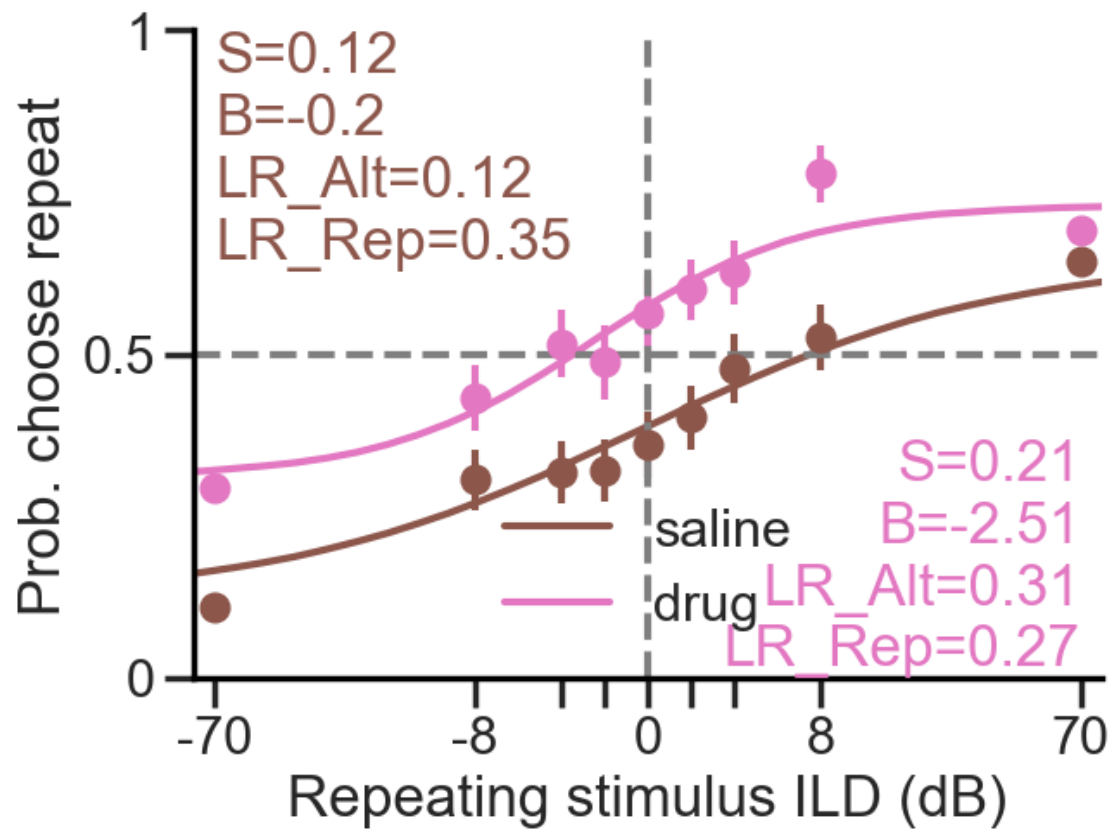


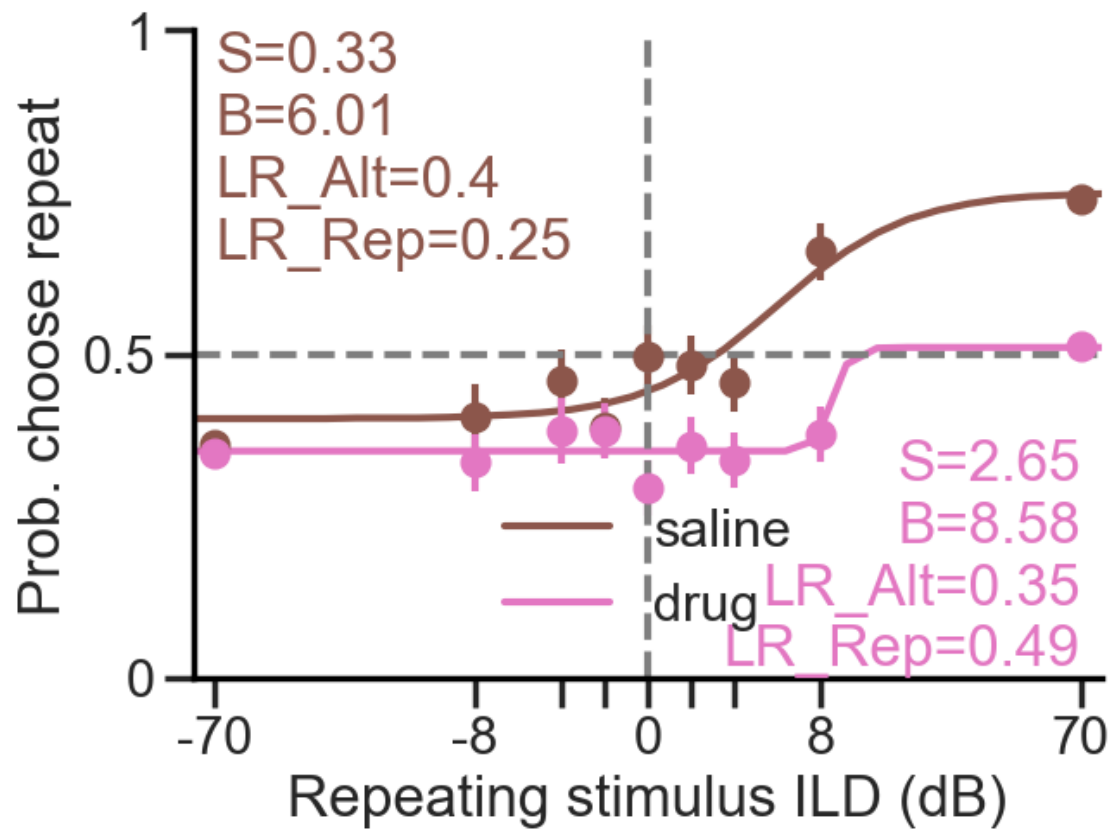


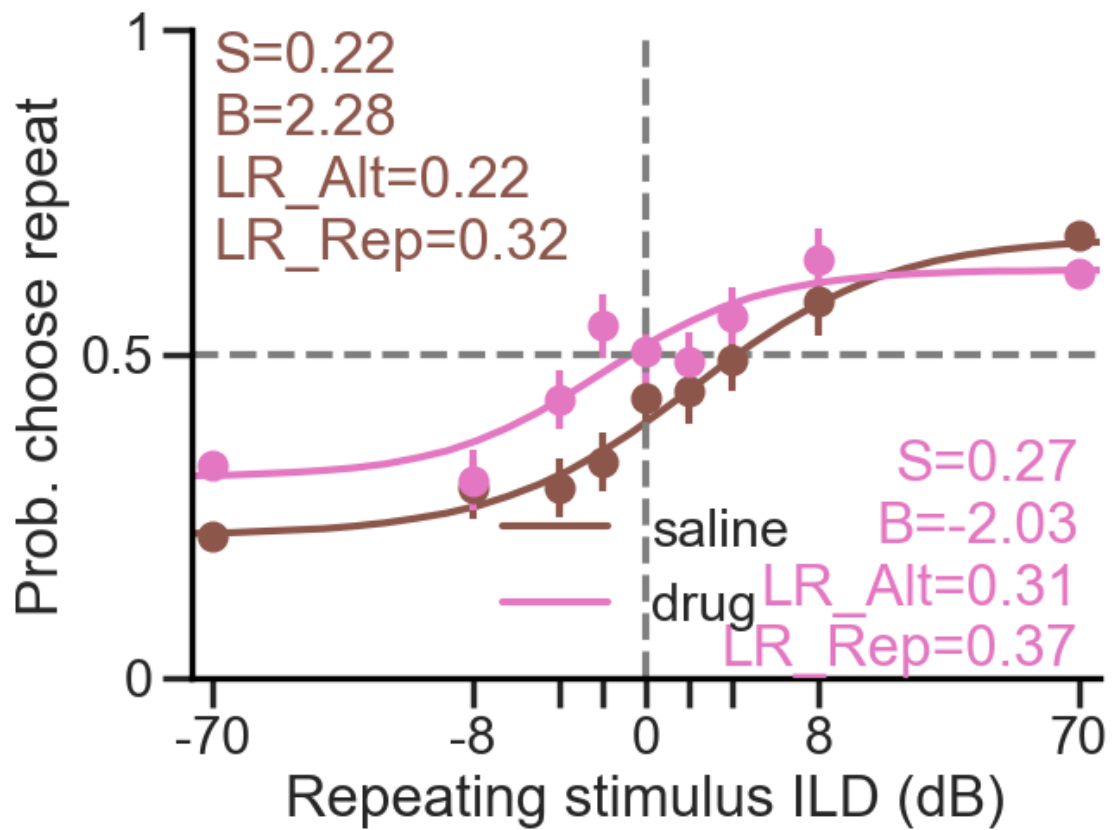


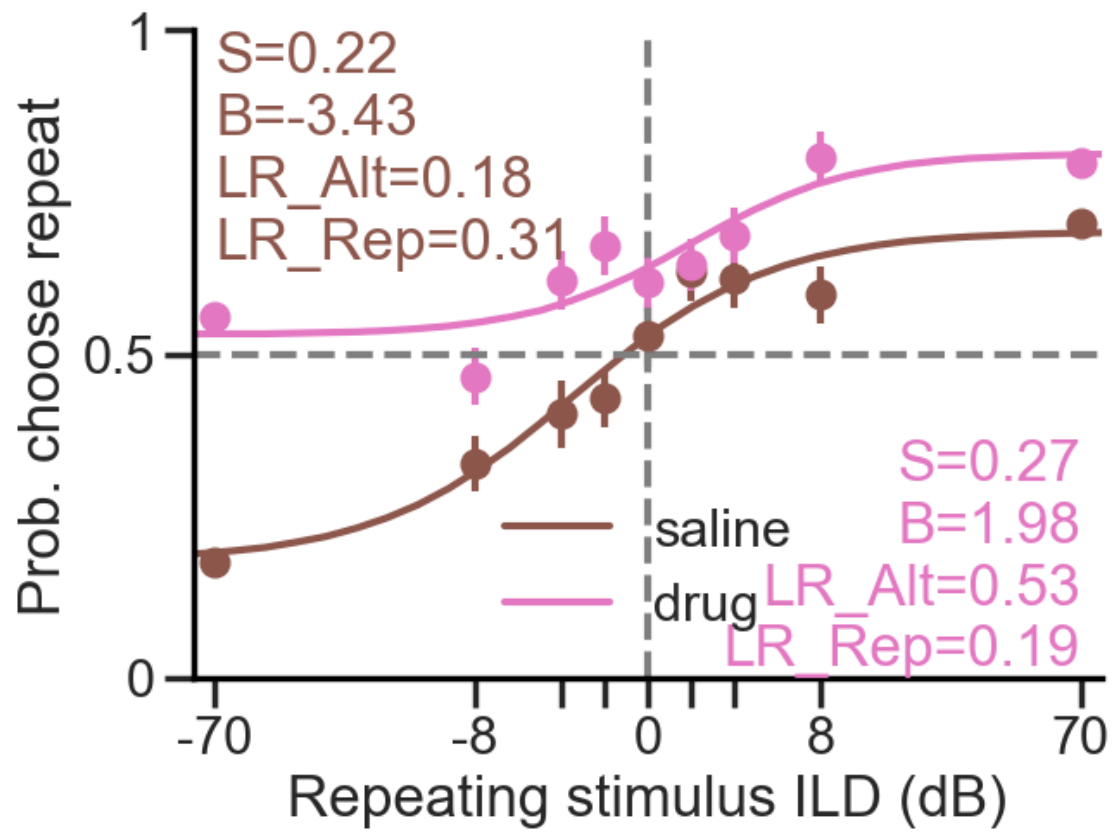




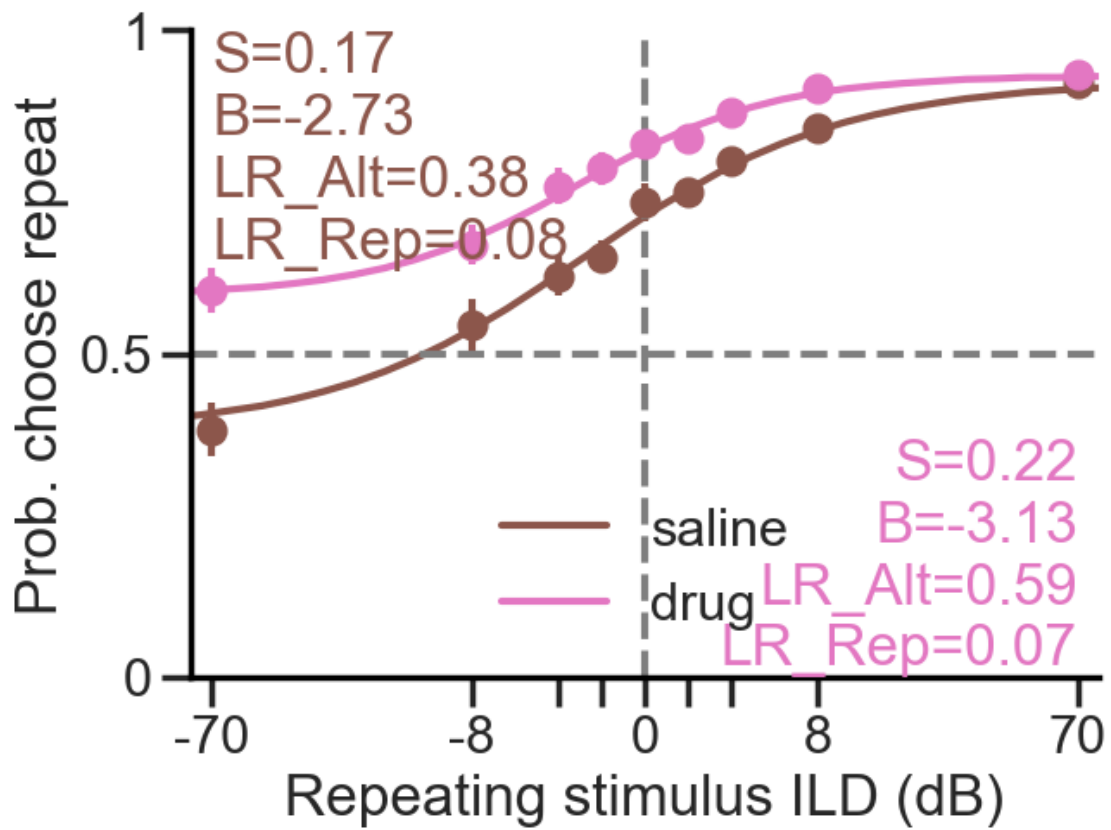






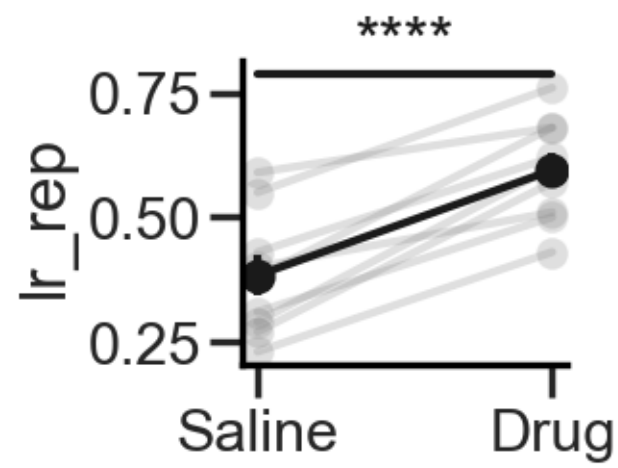
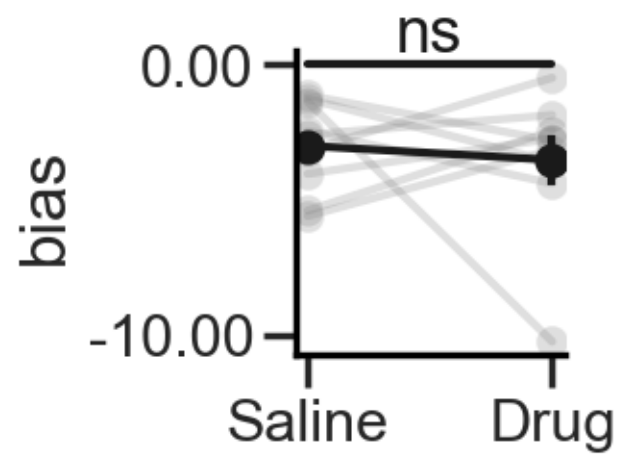
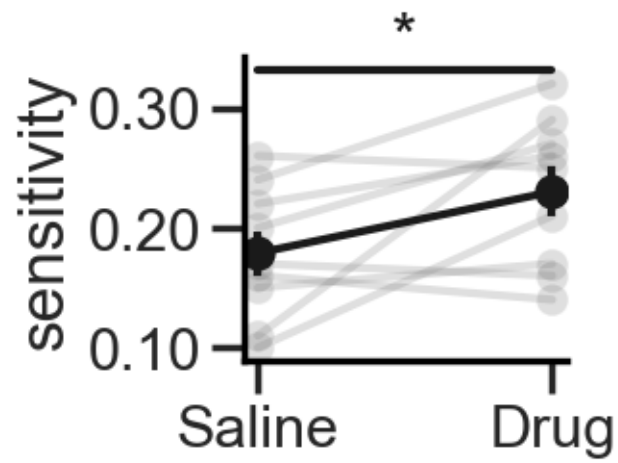


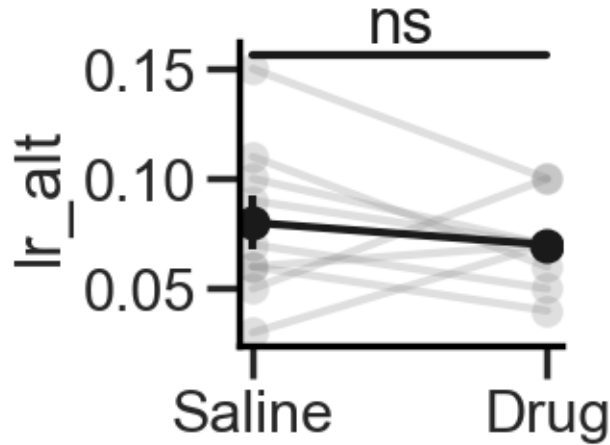
6.2.2 Mean across animals



Stats

sensitivity: t-statistic = -2.322, p-value = 0.049
bias: t-statistic = 0.419, p-value = 0.686
lr_rep: t-statistic = -7.894, p-value = 0.000
lr_alt: t-statistic = 0.832, p-value = 0.430





7 Kernels

7.1 Psychophysical kernels

7.1.1 Mean across animals

2AFC_6

C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6

Getting kernel of animal 014 (1/9)

get_pk took 10.23 seconds to run

Getting kernel of animal 016 (2/9)

get_pk took 15.02 seconds to run

Getting kernel of animal 017 (3/9)

get_pk took 15.24 seconds to run

Getting kernel of animal 020 (4/9)

get_pk took 23.65 seconds to run

Getting kernel of animal 021 (5/9)

get_pk took 14.22 seconds to run

Getting kernel of animal 022 (6/9)

get_pk took 9.48 seconds to run

Getting kernel of animal 023 (7/9)

get_pk took 9.20 seconds to run

Getting kernel of animal 024 (8/9)

get_pk took 12.19 seconds to run

Getting kernel of animal 025 (9/9)

get_pk took 14.49 seconds to run

get_mean_pk took 123.95 seconds to run

2AFC_6

C:\Users\Usuario\PycharmProjects\glue_sessions\2AFC_6

Getting kernel of animal 014 (1/9)

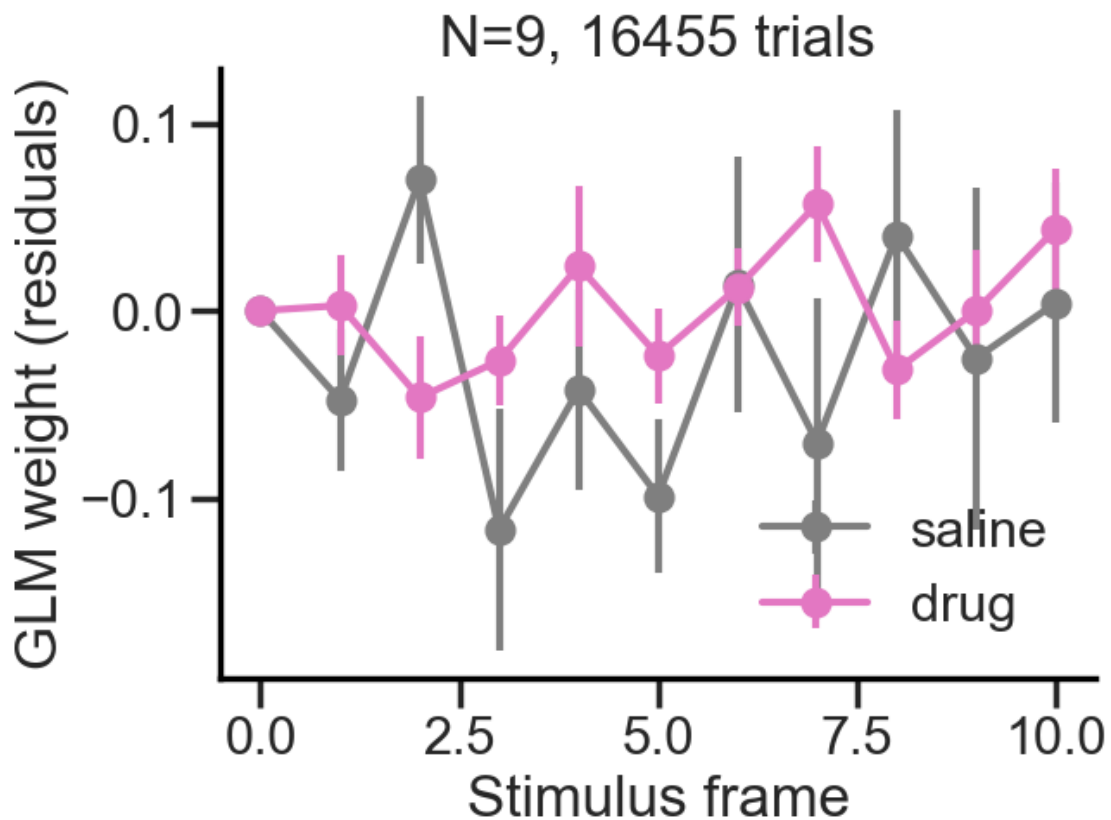
```

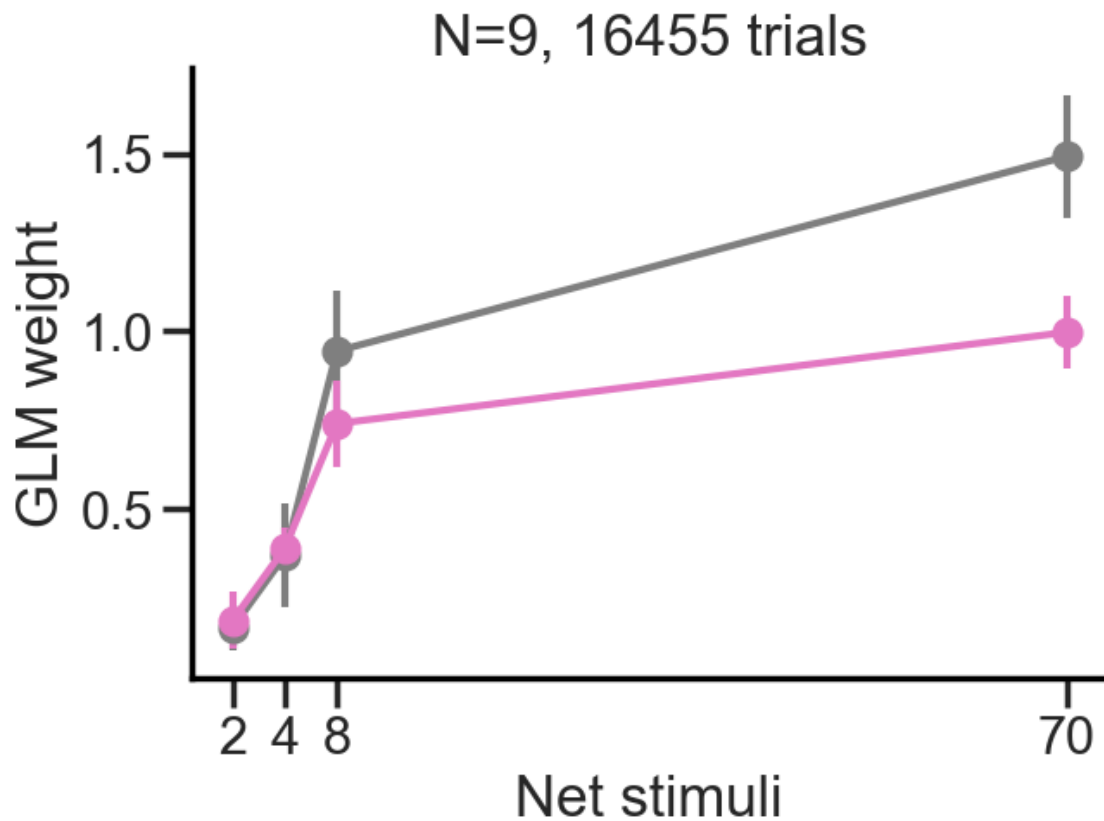
get_pk took 11.91 seconds to run
Getting kernel of animal 016 (2/9)
get_pk took 14.15 seconds to run
Getting kernel of animal 017 (3/9)
get_pk took 14.15 seconds to run
Getting kernel of animal 020 (4/9)
get_pk took 21.89 seconds to run
Getting kernel of animal 021 (5/9)
get_pk took 13.77 seconds to run
Getting kernel of animal 022 (6/9)
get_pk took 14.02 seconds to run
Getting kernel of animal 023 (7/9)
get_pk took 11.65 seconds to run
Getting kernel of animal 024 (8/9)
get_pk took 12.28 seconds to run
Getting kernel of animal 025 (9/9)
get_pk took 15.19 seconds to run
get_mean_pk took 129.25 seconds to run

```

No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored when legend() is called with no argument.

```
plot_drug_pk took 253.23 seconds to run
```





7.2 History kernels

8 Serial dependence

Serial dependence requires a continuous readout of behavioral responses. As choices are binary (left/right), other readouts are needed. Let's try with reaction times (RT) and lick rate (nLicks)

	Trial	Side	PreviousSide	RepSide	RepTrial	Reward	Punish	Miss	\	
0	153	1	NaN	0	1.0	1	0	0		
1	152	1	1.0	1	1.0	1	0	0		
2	151	1	1.0	1	1.0	1	0	0		
3	150	1	1.0	1	0.0	1	0	0		
4	146	1	1.0	1	1.0	1	0	0		
	WrongLick	Hit	...	LicksRight						\
0	0	1.0	...	[1.9182000000000001, 2.0352, 2.1423, 2.2552999...						
1	0	1.0	...	[1.9499000000000004, 2.067, 2.1710000000000003...						
2	0	1.0	...	[1.9076999999999997, 2.0181, 2.138699999999999...						
3	0	1.0	...	[1.9677000000000002, 2.108, 2.22, 2.3375999999...						

```
4          0  1.0  ...  [1.943, 2.0660000000000003, 2.1828, 2.29449999...
```

	nLicksLeft	nLicksRight	nLicks	leftILI	rightILI	ILI	RT \
0	0	8	1.339193	NaN	0.125586	0.125586	0.087397
1	0	8	1.339193	NaN	0.122971	0.122971	0.119097
2	0	4	-2.660807	NaN	0.121300	0.121300	0.076897
3	0	6	-0.660807	NaN	0.126860	0.126860	0.136897
4	0	7	0.339193	NaN	0.120250	0.120250	0.112197

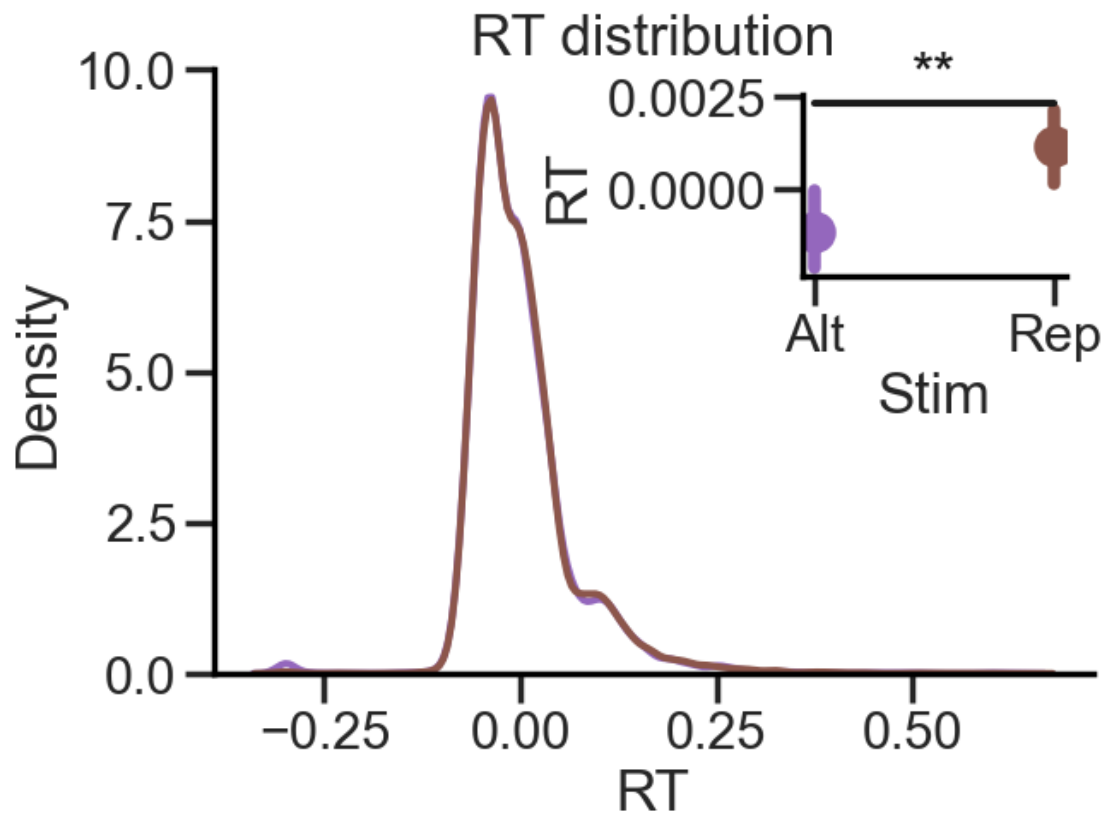
	RT2	SessionHalf
0	0.087397	1
1	0.119097	1
2	0.076897	1
3	0.136897	1
4	0.112197	1

```
[5 rows x 96 columns]
```

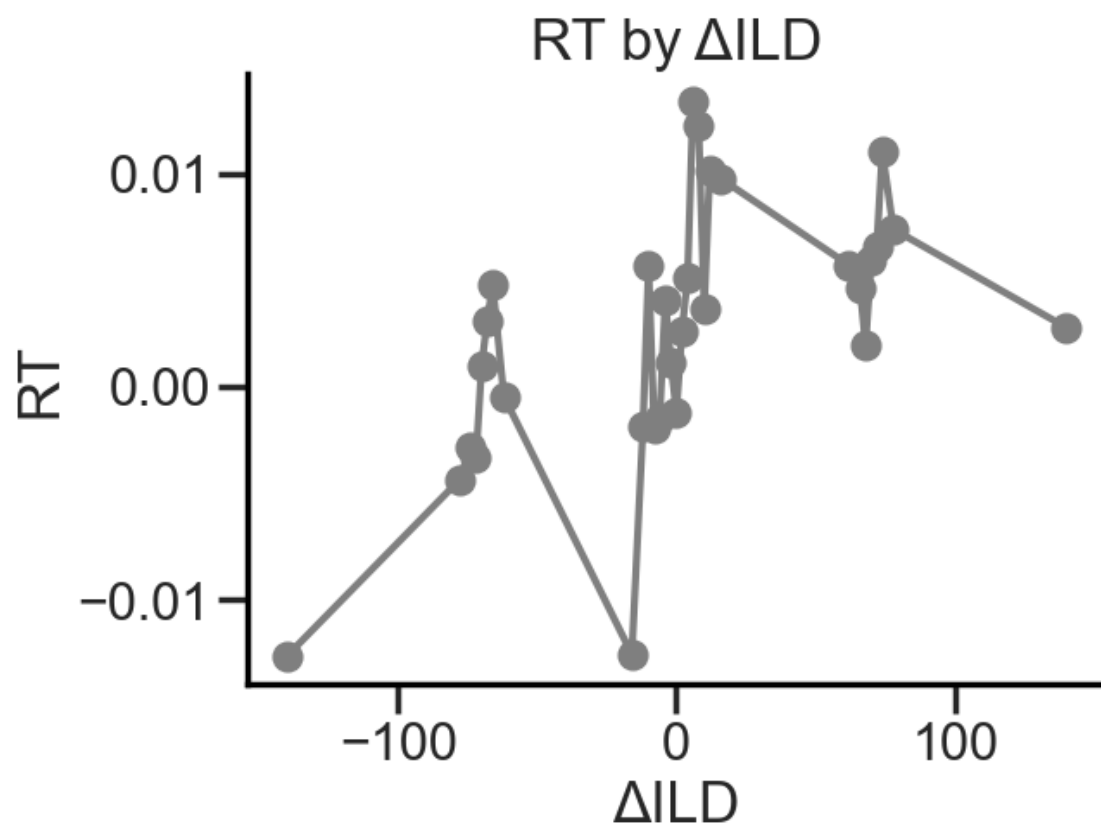
```
array([ nan, -140., -78., -74., -72., -70., -68., -66., -62.,
       -16., -12., -10., -8., -6., -4., -2.,  0.,  2.,
         4.,  6.,  8., 10., 12., 16., 62., 66., 68.,
        70., 72., 74., 78., 140.] )
```

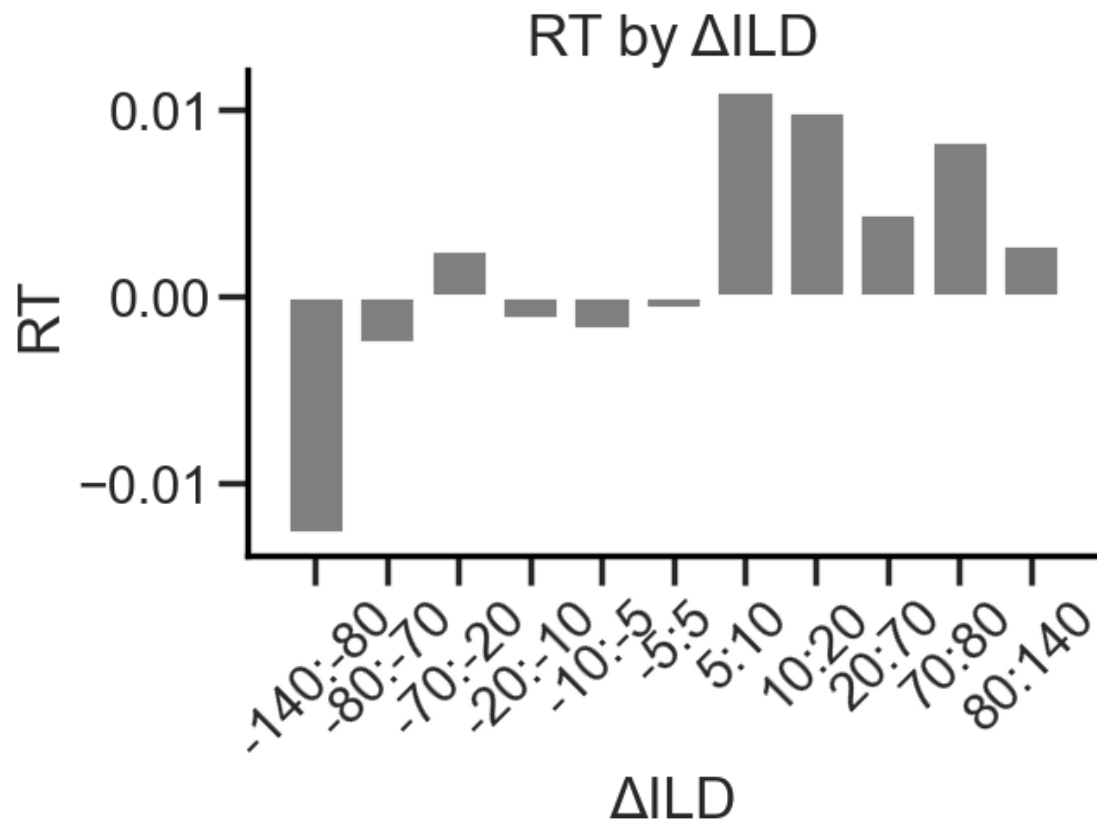
8.0.1 Measured as Reaction Times (RTs)

RT: t-statistic = -2.968, p-value = 0.003

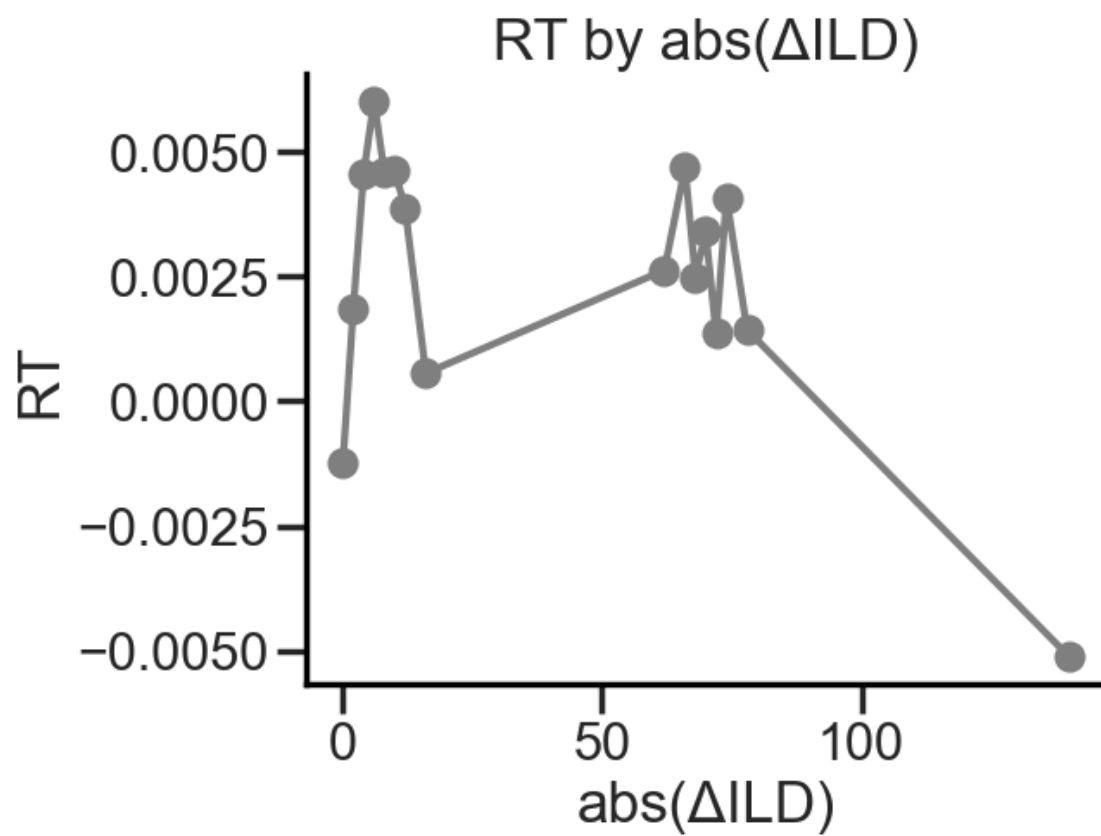


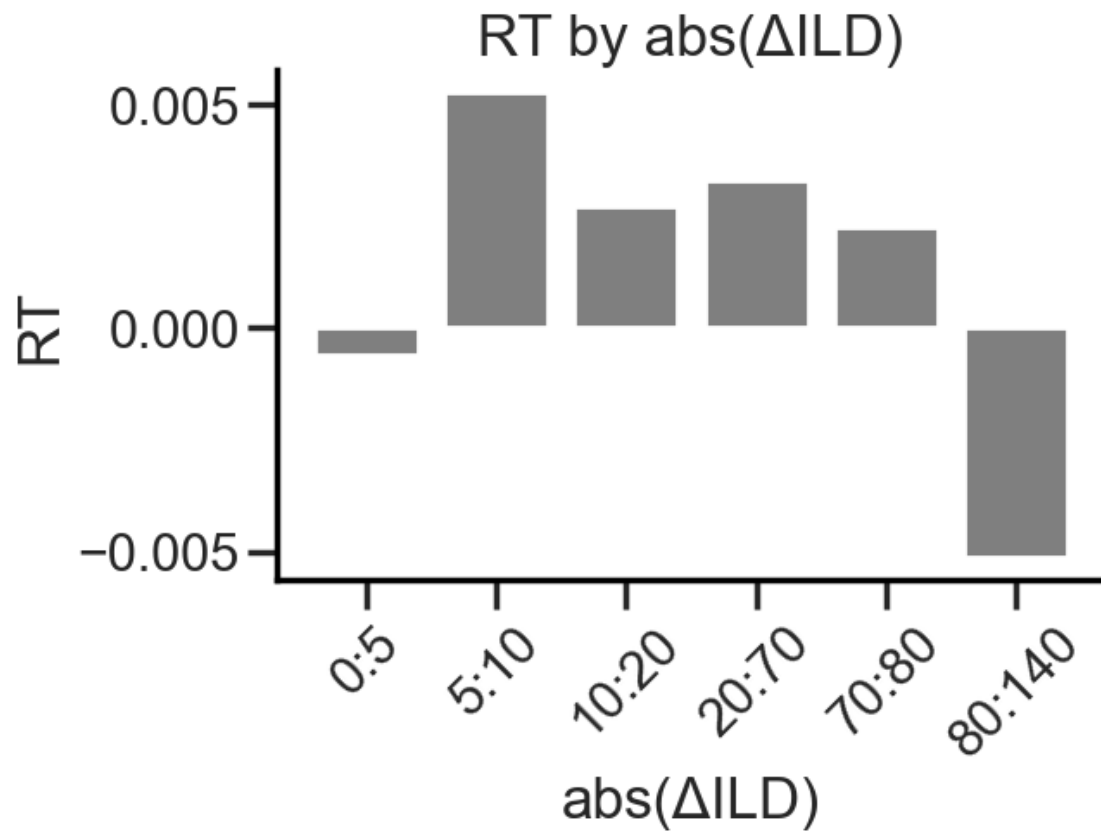
Δ ILD





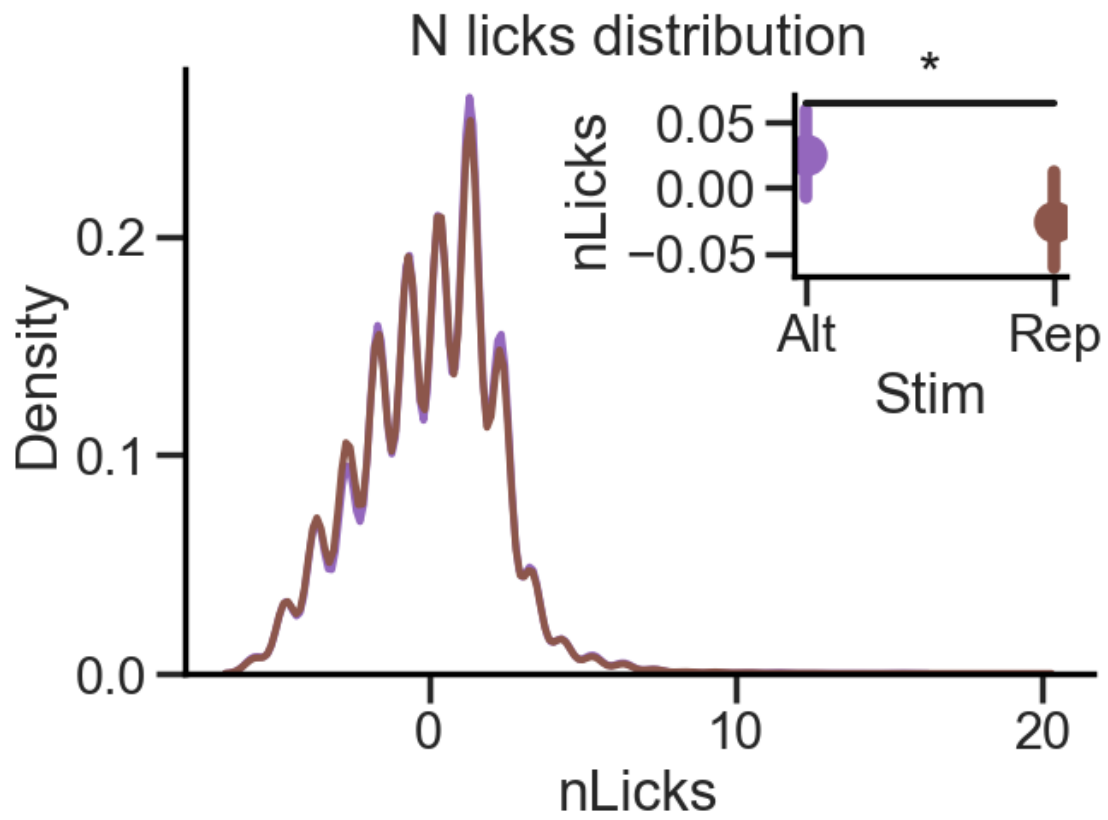
Absolute Δ ILD



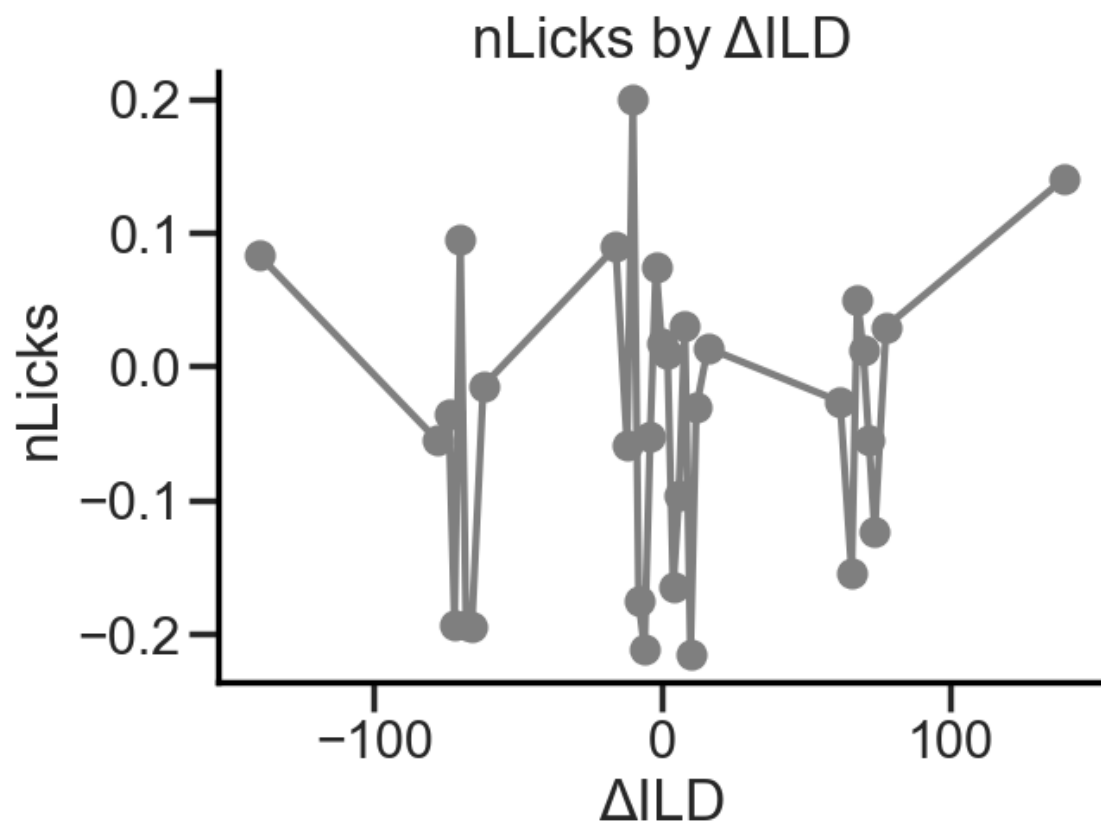


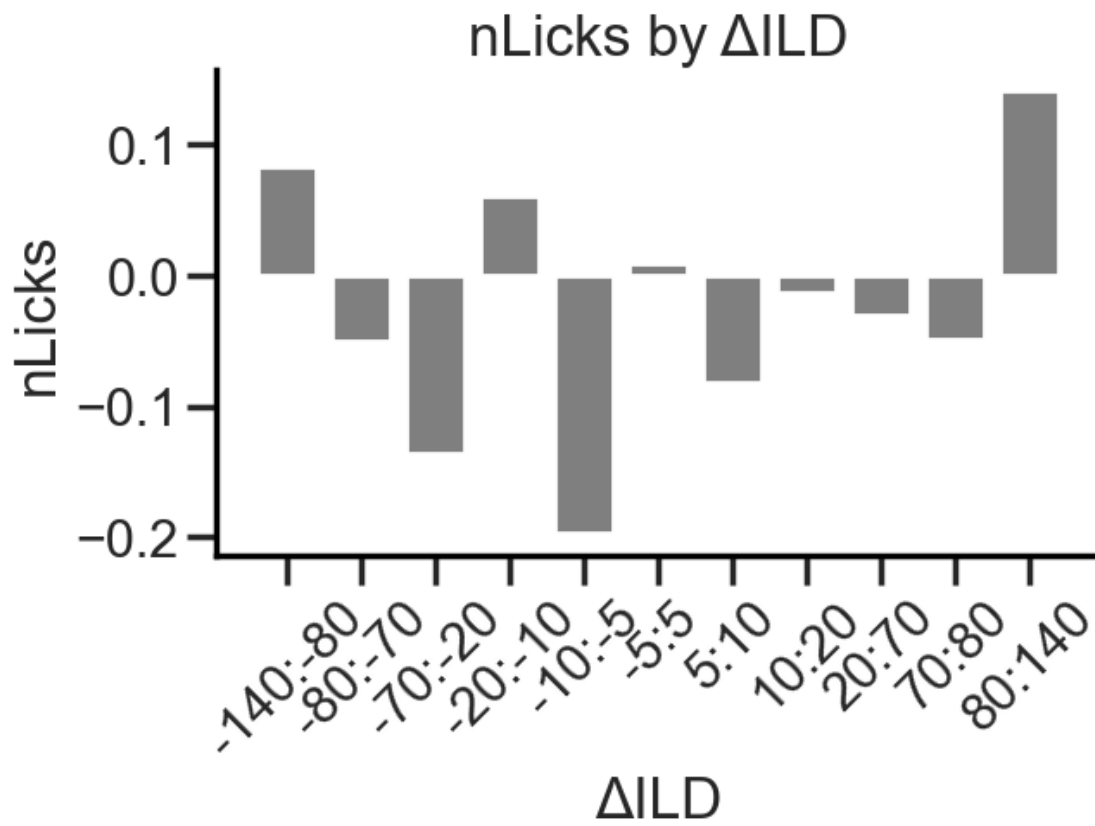
8.0.2 Measured as N licks

nLicks: t-statistic = 2.009, p-value = 0.045

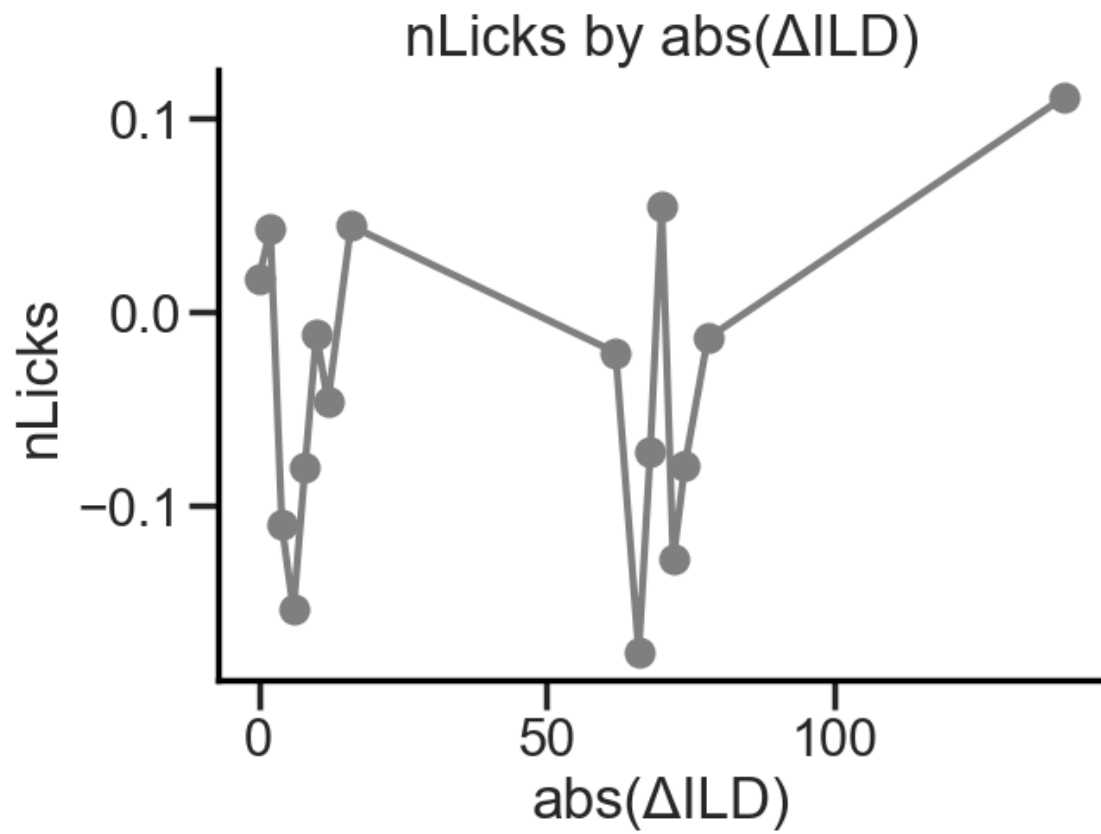


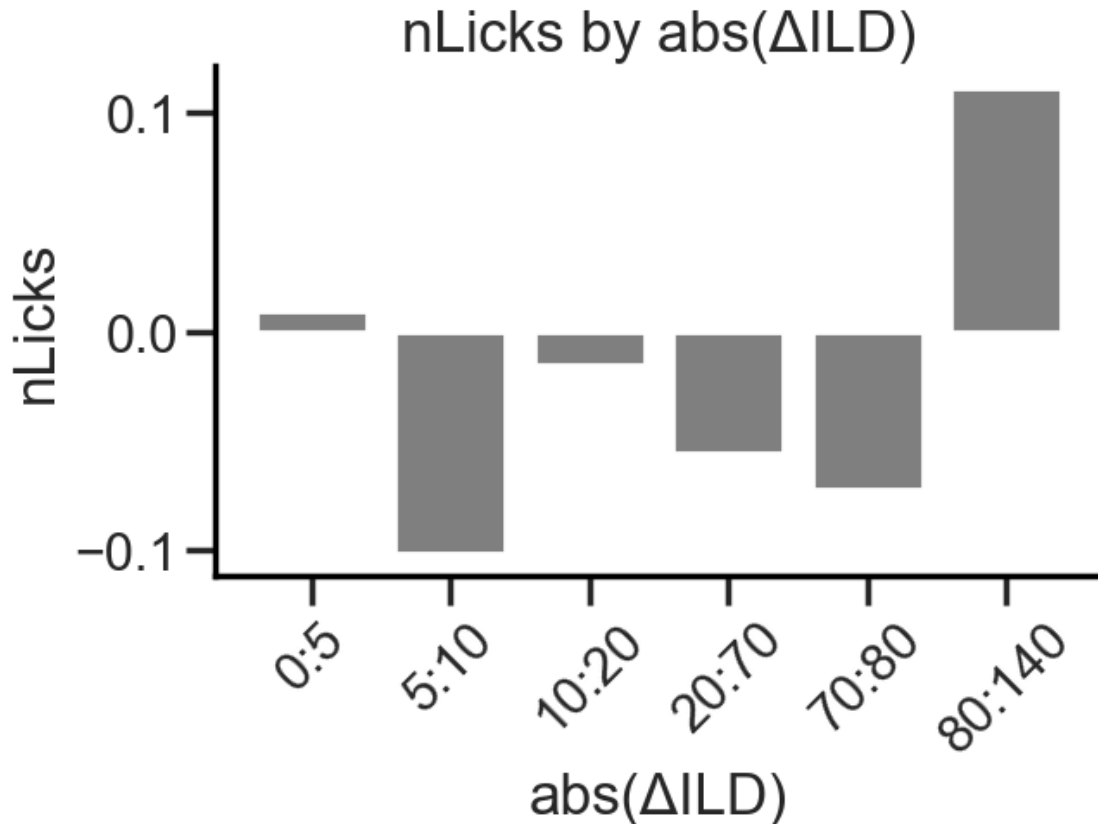
Δ ILD





Absolute Δ ILD





9 Save all notebook's figures

```
[NbConvertApp] Converting notebook notebook.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 52 image(s).
[NbConvertApp] Writing 3638050 bytes to notebook.html

Saved 52 images to notebook_files/

[NbConvertApp] Converting notebook notebook.ipynb to pdf
[NbConvertApp] Support files will be in notebook_files\
[NbConvertApp] Making directory .\notebook_files
[NbConvertApp] Writing 37012 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | b had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 2225240 bytes to notebook.pdf
Traceback (most recent call last):
  File "C:\Users\Usuario\anaconda3\envs\PycharmProjects\Scripts\jupyter-
```

```

nbconvert-script.py", line 10, in <module>
    sys.exit(main())
  File "C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-
packages\jupyter_core\application.py", line 283, in launch_instance
    super().launch_instance(argv=argv, **kwargs)
  File "C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-
packages\traitlets\config\application.py", line 1075, in launch_instance
    app.start()
  File "C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-
packages\nbconvert\nbconvertapp.py", line 420, in start
    self.convert_notebooks()
  File "C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-
packages\nbconvert\nbconvertapp.py", line 597, in convert_notebooks
    self.convert_single_notebook(notebook_filename)
  File "C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-
packages\nbconvert\nbconvertapp.py", line 566, in convert_single_notebook
    write_results = self.write_single_notebook(output, resources)
  File "C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-
packages\nbconvert\nbconvertapp.py", line 526, in write_single_notebook
    return self.writer.write(output, resources, notebook_name=notebook_name)
  File "C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-
packages\nbconvert\writers\files.py", line 155, in write
    with open(dest_path, "wb") as f:
PermissionError: [Errno 13] Permission denied: 'notebook.pdf'

```